

THE  TIMES

Includes
Super Fiendish

Su Doku

The original, best-selling Su Doku series

Book

9

2	5	4	7	3	6	1	8	9
3	9	1	2	8	5	4	6	7
8	7	6	1	9	4	2	3	5
4	1	2	9	6	8	7	5	3
9	6	3	4	5	7	8	2	1
5	8	7	3	1	2	6	9	4
7	2	9	8	4	3	5	1	6
6	3	8	5	7	1	9	4	2
1	4	5	6	2	9	3	7	8

5 million copies sold worldwide

THE  TIMES

Su Doku

Book 9

Compiled by sudokusolver.com

First published in 2009 by Times Books

Collins, an imprint of
HarperCollins Publishers
77–85 Fulham Palace Road
London
W6 8JB

www.collins.co.uk

© 2009 HarperCollins Publishers

All puzzles supplied by Lydia Ade and Noah Hearle of
sudokusolver.com

10 9 8 7 6 5 4 3 2

The Times is a registered trademark of Times Newspapers Ltd

ISBN 978-0-00-730584-1

All rights reserved. No part of this publication may be reproduced, stored in a retrieval system or transmitted in any form or by any means, except with the prior written permission of the publisher and copyright owner.

A catalogue record for this book is available from the British Library.

Layout by Susie Bell

Printed and bound in Great Britain by Clays Ltd, St Ives plc



Mixed Sources

Product group from well-managed
forests and other controlled sources
www.fsc.org Cert no. SW-COC-1806
© 1996 Forest Stewardship Council

FSC is a non-profit international organisation established to promote the responsible management of the world's forests. Products carrying the FSC label are independently certified to assure consumers that they come from forests that are managed to meet the social, economic and ecological needs of present and future generations.

Find out more about HarperCollins and the environment at

www.harpercollins.co.uk/green

Contents

Introduction iv

Puzzles

Mild	1-10
Difficult	11-70
Fiendish	71-120
Super Fiendish	121-150

Solutions

Introduction

Welcome to *The Times Su Doku Book 9*. This book contains Mild, Difficult, Fiendish and Super Fiendish puzzles, sorted within each rating in order of increasing difficulty.

This introduction covers the harder solving techniques that you will need to solve Fiendish and Super Fiendish Su Doku puzzles. If you do not already know the easier solving techniques, they are covered in many of our other Su Doku books.

Fig. 1 shows a partially solved Fiendish Su Doku puzzle. Since no more squares can be solved with easy solving techniques, full pencil marks have been made.

Fig. 1

7	1	249	8	5	39	49	234	6
5	269	269	39	7	4	8	23	1
8	3	49	2	1	6	4579	47	579
16	4	5	69	3	179	2	8	79
9	268	3	56	68	257	57	1	4
12	28	7	4	89	1259	3	6	59
126	269	1269	7	469	8	146	5	3
3	5679	8	1	469	59	467	47	2
4	567	16	356	2	35	167	9	8

Naked Pair

Two squares in the eighth column both contain only {4 7} (circled in Fig. 2). Therefore, you know that these two numbers must be placed in some order in these two squares, and not in any other in the column. You can now remove {4 7} from all other pencil marks in this column, as shown in Fig. 2.

Fig. 2

7	1		8	5			²³⁴	6
5				7	4	8	²³	1
8	3		2	1	6		⁴⁷	
	4	5		3		2	8	
9		3					1	4
		7	4			3	6	
			7		8		5	3
3		8	1				⁴⁷	2
4				2			9	8

Hidden Pair

In the second column, although it may not be immediately obvious, the pencil marks {5 7} only appear in two squares in this column (see Fig. 1). Therefore, these two numbers must be placed in some order in these two squares, and you can now remove all other possibilities from these squares, as shown in Fig. 3, overleaf.

Fig. 3

7	1		8	5				6
5	²⁶⁹			7	4	8		1
8	3		2	1	6			
	4	5		3		2	8	
9	²⁶⁸	3					1	4
	²⁸	7	4			3	6	
	²⁶⁹		7		8		5	3
3	5	8	1					2
4	5			2			9	8

Usually, using these methods will help you to progress through the puzzle, although in this example, they have not helped to solve any squares so far. In this case, you will need to look for naked triples.

Naked Triple

In the central 3x3 block, there are three squares which all only contain some of the numbers {6 8 9}. Using the same logic as the naked pair, these three numbers must be placed in some order in these three squares. Therefore, you can remove {6 8 9} from the pencil marks of all other squares in this block, as shown in Fig. 4.

You can now solve the 5 in this block. After this, the rest of this Su Doku can be solved without difficulty.

Fig. 4

7	1		8	5				6
5				7	4	8		1
8	3		2	1	6			
	4	5	6 9	3	1 7	2	8	
9		3	5 8	6 8	257		1	4
		7	4	8 9	1 25	3	6	
			7		8		5	3
3		8	1					2
4				2			9	8

Take a look at the Super Fiendish Su Doku puzzle in Fig. 5. Again, this puzzle cannot be solved any further with easy solving techniques, and full pencil marks are shown.

Fig. 5

1	25679	56789	3	2789	278	257	2579	4
2579	3	579	17	1279	4	257	8	6
2789	279	4	6	2789	5	1	3	29
4	1	3	2	57	6	8	579	59
5789	5679	56789	178	3	178	4	2579	259
578	57	2	4	578	9	6	1	3
27	247	1	5	247	3	9	6	8
3	8	57	9	6	27	25	4	1
6	2459	59	18	1248	128	3	25	7

X-wing

Look closely at the 8 pencil marks in the third and sixth rows. Each row contains only two squares with the 8 pencil mark (highlighted in Fig. 6), and in both rows these two squares lie in the first and fifth columns.

Fig. 6

1			3					4
	3				4		8	6
2 ²⁷ 8 ⁹	2 ⁷ 9	4	6	2 ⁷ 9	5	1	3	2 ⁹
4	1	3	2		6	8		
				3		4		
5 ⁷ 8	5 ⁷	2	4	5 ⁷ 8	9	6	1	3
		1	5		3	9	6	8
3	8		9	6			4	1
6						3		7

There are only two possible cases for the placing of the 8 in the third row. Firstly, if it is placed in the first column, then the 8 for the sixth row must be placed in the fifth column. Secondly, if the 8 is placed in the fifth column, then the 8 for the sixth row has to be placed in the first column. Either way, there will always be one 8 in the first column, and one in the fifth. Therefore, you can eliminate 8 as a possibility from all other squares in these two columns, as shown in Fig. 7.

In general, an x-wing is found when for two rows, there are two, and only two, possible squares in which a particular number can be placed, and for both rows these squares lie in the same two columns. In this case, this particular number can be eliminated as a possibility for all other squares in those two columns. You can exchange rows and columns in this definition.

Fig. 7

1			3	2789				4
2579	3			1279	4		8	6
2789	279	4	6	2789	5	1	3	29
4	1	3	2	57	6	8		
5789				3		4		
578	57	2	4	578	9	6	1	3
27		1	5	247	3	9	6	8
3	8		9	6			4	1
6				1248		3		7

As the x-wing method did not solve any squares, you will need to try another method along the same lines.

Swordfish

In each of the third, sixth and seventh rows, the 7 pencil mark appears in only three squares, highlighted in Fig. 8, overleaf.

Fig. 8

1			3					4
	3				4		8	6
1 ² 3 ⁹	1 ⁷	4	6	1 ² 3 ⁹	5	1	3	2 ⁹
4	1	3	2		6	8		
				3		4		
3 ⁷	5 ⁷	2	4	3 ⁷	9	6	1	3
1 ⁷	2 ⁴ 7	1	5	2 ⁴ 7	3	9	6	8
3	8		9	6			4	1
6						3		7

Additionally, all of these squares lie in only three columns. Using similar logic to the x-wing, whichever way these three 7s are eventually placed in these three rows, there will be one 7 in each column. As before, you can now eliminate the 7 pencil marks from all other squares in these three columns, as shown in Fig. 9.

You can now place the 5 in the fifth column. After this, you can solve the rest of the puzzle using easy techniques.

In general, a swordfish is found when for three rows, there are two or three possible squares in which a particular number can be placed, and for all three rows these squares lie in the same three columns. In this case, this particular number can be eliminated as a possibility from all other squares in those three columns. Again, you can exchange rows and columns in this definition.

Fig. 9

1	25679		3	279				4
2579	3			1279	4		8	6
1089	109	4	6	1089	5	1	3	29
4	1	3	2	57	6	8		
579	5679			3		4		
908	90	2	4	908	9	6	1	3
20	240	1	5	240	3	9	6	8
3	8		9	6			4	1
6	2459			124		3		7

We have more Su Doku solving techniques on our website. So, if you are finding it difficult to progress through any puzzle here, our details are at the back of this book.

Have fun and enjoy the Su Doku!

Lydia Ade and Noah Hearle, sudokusolver.com

Puzzles

Mild

		1		7		5	6	9
3	4	7	5	9	6	1	2	8
5	6	9	8	2	1	7	3	4
	3			5		2	7	
	5		7		2		4	
		4		6			8	5
6		2			7	4	5	3
	1	5	2	3		6	9	7
		3	6		5	8	1	2

3	9			4			6	8
5						4		9
	6	8	9			7		
			8		9	2		
6								7
		9	7		3			
		4			6	3	9	
8		1						4
9	3			2			5	1

	1		6			4	5	
4			5	2				3
5			8		4			
		1				8	6	7
	5						2	
6	4	8				3		
			7		9			2
9				6	3			1
	6	2			8		3	

					7	9	6	
6			1	3				
7		8	5			2		
2				1		8	5	
	8		6		5		9	
	9	5		4				7
		4			8	1		6
				2	1			9
	7	2	9					

3		7	6				5	9
6		8		9	3			
					2		3	8
	3	9	2		4			5
	7						2	
2			8		7	9	6	
5	6		9					
			3	2		4		6
4	8				6	5		3

7							9	4
4	1		2	5				
			3	9		5		
1						6	2	
	9						7	
	2	3						5
		4		8	2			
				6	5		3	8
9	6							2

7					9		1	6
3					5			
			3	1				
1	9			3		5		
		5	1		8	9		
		6		9			7	1
				2	1			
			9					3
4	6		7					2

5			9	7		4		
7				2	4			5
3							8	
						2		
8		1	2		9	5		6
		4						
	8							9
9			3	5				7
		5		9	8			3

	5		4		7		2	
	8			6			4	
4		9				3		6
		4		7		6		
1				2				8
		7		9		1		
6		2				4		7
	3			4			6	
	4		1		6		9	

1	5				6			8
8					7	2		5
					3	1		9
4		1		6		8		7
7		3	2					
9		4	1					3
3			5				6	4

NOT SO
Difficult

2	5	3	1	9	4	6	7	8
7	4	1	8	3	6	9	2	5
9	8	6	2	5	7	4	1	3
6	4 7	2	4	8	9	3	5	1
4	3	5	6	7	1	8	9	2
1	9	8	5	2	3	7	6	4
8	6	9	3	1	2	5	4	7
3	1	7	9	4	5	2	8	6
5	2	4	7	6	8	1	3	9

5				9	1			
			2					
			5	7	3	2		
	8	5				9		2
2		4		6		7		5
9		7				3	1	
		8	9	3	5			
					6			
			1	4				6

					1	5	6	2
	2				9			4
		6				8		7
				7			4	6
			1		8			
2	3			5				
3		4				6		
6			4				5	
8	9	1	5					

			3			2		
	2		5		8	7	4	
8	9		6					
	4					9	5	8
1	8	6					7	
					4		2	3
	5	4	8		3		6	
		8			1			

6		3	4		9		8	
							7	3
2				1		9		
5			9		6			8
		8				7		
4			8		1			5
		4		6				9
8	2							
	9		5		3	6		4

	5	2	6					
				9	8			4
		6				1		3
	4			2				5
	6		7		3		9	
8				5			1	
3		9				2		
6			3	4				
					2	8	3	

	1		4	6			5	
2							1	6
					8	3		
9				7		8		
8			1		3			5
		7		8				2
		8	7					
1	4							7
	9			2	1		6	

8		6			9			
	5			3		8		
		3	5					4
		7		5			9	
			3		4			
	8			2		5		
4					1	9		
		5		6			4	
			8			1		6

5		2						4
	6			2			8	
			5		3			7
		3	4		1	8		
	9						1	
		6	2		7	5		
7			8		5			
	2			7			5	
6						7		3

7			6	8	3			4
6	9			2			7	8
			9		7			
		6				4		
	2			6			9	
		3				8		
			8		1			
3	5			7			8	2
1			3	9	2			7

			2		9	3		4
2		4	1					
3		9						
4							8	7
			6		8			
1	6							2
						1		6
					4	2		8
8		2	5		3			

					3	1	9	
8						4		
5	3	4			1	6		
7		9	5		6			
			4		7	9		3
		3	2			5	4	1
		5						6
	8	2	6					

	2			8			6	
	9			5			3	
		4	1		7	8		
5								3
	8	2	5		3	9	7	
1								4
		8	3		2	6		
	6			7			4	
	1			6			2	

	8	6		9		3	7	
4				3				6
			4		7			
		8				2		
3	4			1			8	9
		7				4		
			6		3			
8				5				7
	6	1		7		9	5	

8	6			7			1	5
5			9	8				6
		2		1				
	5							
3	2	7				8	5	1
							3	
				4		9		
1				9	6			7
7	9			5			4	3

5			6	2	9			
				4				3
1				7			9	5
		2					8	
8	7						3	9
	1					7		
3	6			9				1
4				6				
			4	3	5			8

	7			8				
	5			1	9		3	6
		3	6			5		
	1					2		
3	9			5			4	8
		8					9	
		1			5	7		
5	4		8	2			6	
				7			5	

		7	6				8	1
								5
8	2							9
		6	8	3			7	
		3				9		
	9			2	7	6		
6							3	2
5								
9	3				2	5		

6	3						1	4
7	4		1		6		5	2
	5			7			9	
			8		5			
	8			3			7	
1	6		4		9		3	8
4	7						2	1

	2	9		3	5	6		
5	3							
8			6					2
		2		5				9
9			8		3			1
3				1		8		
7					1			3
							7	6
		3	7	4		1	9	

9		1		6				4
		6	1					
				7	2		3	1
		8					5	
4		3				7		2
	9					1		
3	6		2	9				
					7	9		
8				5		2		6

5	8							4
		9	3	5				8
		6				5	3	
			6		8		1	
	1			3			2	
	3		2		1			
	4	7				2		
3				1	9	8		
8							7	3

			6		1	4		9
							5	
				4	8	6		3
9			8			2		5
		4		1		8		
8		7			3			6
6		2	3	7				
	7							
3		1	5		6			

		4				9	2	
6				4	2			
5			6					7
	1		8		5	7		
	5			2			9	
		7	4		3		8	
7					6			9
			5	3				4
	8	3				1		

	3		7		2		9	
2	4						7	5
				3	8			
9		7						1
		6		8		5		
5						7		6
			8	2				
8	9						1	4
	5		6		4		8	

3		9		7		1		
	2			5				
								6
	4				7			9
		5	1		6	3		
7			3				2	
1								
				6			8	
		6		3		9		5

4	3	9	2					5
	5		3				2	6
					7			8
		6					9	1
5	2					8		
1			4					
2	6				3		5	
8					1	2	6	9

	1	7			4			5
8							4	
3		2	5			9		
		4		8	5			6
			4		6			
7			3	9		8		
		8			1	7		2
	2							8
6			8			4	3	

6			9				8	2
5			8	4				
					3			
		6		9			1	7
	9		2		8		4	
4	8			7		6		
			7					
				3	6			1
9	3				5			6

					2		4	
1	9	4					2	
		5	4			3	8	
6			8		5	4		
		8	3		1			2
	4	2			3	8		
	8					2	7	1
	5		2					

		2	5			8		7
	1					3		6
3				8	9		1	
7		6	9					1
1					6	2		4
	4		7	9				8
9		1					4	
2		5			8	6		

6					2			7
				7		8		
	7	5	1			4		
5			2		1	7		
	3			5			6	
		7	8		6			1
		2			4	6	7	
		1		9				
4			7					8

	2	8	6				5	
	5		7				6	
				5	9	3		
8			5					
9			3	6	7			1
					8			6
		2	9	3				
	9				6		7	
	8				2	4	9	

	2			1				9
		3		2	5	8	1	
8	4		3					
			7			9		
	7			4			6	
		8			3			
					7		4	5
	3	7	5	8		2		
2				3			8	

			4		2			
				8		4		6
		3		6		7		
3				4		9		5
	7		8		5		6	
5		2		7				3
		4		1		5		
6		1		5				
			3		8			

2		4				1		9
		1				4		
8				9				5
			9		8			
9		3	4		7	8		1
			3		2			
7				4				6
		8				9		
1		5				3		8

		5		6	7		2	
4			5					
				2	3			4
6		8					9	
2		9		4		7		1
	3					4		6
3			2	8				
					6			3
	9		7	3		2		

2			5		9		3	6
1			7		2			
		3		1		2		
3	5						9	8
		8				6		
4	6						1	3
		1		4		3		
			8		3			1
8	3		1		5			2

7	1					2		8
		2			1			3
5					9			
	6		1	4		3	7	
	4	1		2	5		8	
			8					7
8			3			5		
1		4					2	9

		5	7		8		1	4
			4		2			
	1							
8		6		4		7		5
7		3		8		6		1
							6	
			9		4			
2	7		5		3	9		

9		3		8		7		1
	4		9		1			
8								9
1		4		9		6		
			1		4			
		9		5		3		4
2								8
			5		2		3	
5		1		3		9		6

	2		6		3		7	1
		4					2	
6		1			4	8		
			9		2	7		
				6				
		2	1		5			
		3	8			1		7
	1					5		
5	8		2		9		3	

		6	8		2	1		
		4	9	1	7	8		
	7		2		5		4	
	8						2	
	9		6		1		8	
		8	7	2	6	4		
		5	1		9	6		

9	6	2	3	7	8	4	1	5
3	5	4	2	9	1	7	8	6
8	1	7	4	6	5	3	2	9
2	9	8	7	4	6	1	5	3
4	7	1	8	5	3	6	9	2
6	3	5	1	2	9	8	7	4
1	2	3	9	8	4	5	6	7
7	4	6	5	1	2	9	3	8
5	8	9	6	3	7	2	4	1

					1			6
2	3		7				1	
	1	4			8		9	
1		8	6	5				
7	9						5	1
				9	7	8		3
	8		3			1	2	
	6				5		8	7
9			8					

				9				
	8		7					9
5					2	8		6
		6		4		7	2	
		4				3		
	3	2		5		6		
2		7	1					5
1					9		3	
				7				

	1				7			
7			4				6	5
		5			8			4
	9			7				
6								2
				2			1	
3			7			5		
4	8				6			1
			8				7	

9	6	1	4	5	3	7	8	2
3	4	5	2	7	8	6	1	9
7	2	8	1	6	9	3	5	4
6	3	4	5	9	2	1	7	8
2	5	7	6	8	1	9	4	3
1	8	9	7	3	4	5	2	6
5	7	3	8	4	6	2	9	1
8	9	2	3	1	7	4	6	5
4	1	6	9	2	5	8	3	7

		4					5	8
	7		4		2			9
2				3				
	6		2		9		8	
		5				9		
	3		8		1		4	
				7				6
9			1		6		3	
6	8					1		

				5		9	1	7
	8		1					
	5		9					6
	1	2	8					
8	6						7	5
					9	2	8	
5					6		2	
					1		3	
7	3	1		4				

	6	7			5			
9	1						6	
4					6	7		
			2		4	1		3
8		1	9		7			
		9	3					2
	2						5	1
			6			8	4	

	5	8	1		3		2	
2							1	
		7	2			8		
						2		
	4		7	2	8		5	
		6						
		3			5	4		
	8							6
	6		8		4	1	9	

			9		3		8	
4								
		7		6	4	2		
5		4	2		6			8
		6				4		
9			8		1	6		7
		9	7	3		8		
								3
	1		6		2			

		8	4				2	5
7					5			
		4		8		6		9
	8		2		3			7
		5				3		
3			1		7		5	
8		3		9		5		
			5					3
	5				1	8		

9		6			7		3	8
		8			3	5		6
		3		4				
								7
8	9						5	4
7								
				5		6		
5		4	9			2		
6	1		3			7		5

				5				4
6	1		9		8			
			7	2		1	9	
7	9							1
2							8	7
	2	7		3	5			
			1		6		2	5
1				7				

4			8				3	
3					1		4	2
	1	6		2				
					7	2		9
				6				
5		3	1					
				5		7	9	
2	9		7					3
	5				3			4

8	4	6	7	9	1	5	2	3
7	1	2	5	4	3	6	9	8
9	3	5	2	6	8	4	7	1
4	7	9	3	2	5	8	1	6
2	5	1	6	8	9	7	3	4
6	8	3	4	7	4	2	5	9
3	6	7	4	1	2	9	8	5
5	2	8	9	3	6	1	4	7
1	9	4	8	5	7	3	6	2

1	3	5	9	8	4	6	2	7
2	6	9	1	5	7	4	8	3
8	4	7	2	6	3	5	9	1
3	9	6	4	7	5	2	1	8
4	7	1	6	2	8	3	5	9
5	8	2	3	1	9	7	6	4
9	5	8	7	4	2	1	3	6
6	2	4	8	3	1	9	7	5
7	1	3	5	9	6	8	4	2

✓ ✓ ✓ ✓
 1 2 3 4 5 6 7 8 9
 —

4	8	7	6	¹⁹ 9	¹⁹ 1	3	5	2
2	3	6	8 5	5 7	1	4	9	
⁵⁰ 9	1	5 5	4	3	2	8	7	6
8	⁶⁹ 9	2	¹⁵ 1	4	¹⁵ 5	²⁶ 6	¹³² 3	7
⁷ 7	5	3	9	2	6	4	8	¹³ 1
1	⁶⁹ 6	4	3	7	8	²⁶ 2	¹⁴⁰ 9	5
³⁵ 5	2	¹⁰⁵ 9	7	8	4	9	6	¹³ 3
6	4	9	5	¹⁹ 1	3	7	2	8
³⁷ 3	7	8	2	6	9	5	¹³ 1	4

Fiendish

1	4	8 ²⁵	3	5 ⁵⁶	2 ²²	9 ²⁵	7	6
3 ²⁷	6 ²⁶	7	8 ¹⁸	9 ⁵⁹	1	2 ²⁸	5 ²⁵	4
9	2 ²⁶	5 ¹⁵	7 ¹⁷	4	6 ²⁵	8 ¹²	3	1 ¹²
4 ¹⁴	9	2 ²³	6	3 ¹²	8	7 ¹²	1 ¹⁴	5
6	5	1	2	7	4	3	9	8
8	7 ²³	3 ²³	5	1 ¹³	9	4 ¹²	6	2 ¹²
7 ¹²	8	6 ¹⁴	1 ¹⁷	2	3 ¹⁵	5 ¹⁵	4 ¹⁴	9 ¹³
5	1 ¹¹	9 ³	4	8	7 ¹⁷	6	2	3 ³⁷
2	3 ¹³	4 ¹⁴	9 ¹⁷	6 ⁶⁷	5	1 ¹⁴	8	7

✓
5678

1	4	2	³⁴ 3	⁶⁸ 8	7	9	³⁶ 6	5
6	9	7	³⁵ 5	¹⁵⁸ 1	2	8	¹³ 3	4
8	3	5	4	¹⁶ 6	9	2	¹⁶ 1	7
²⁴ 2	7	9	²⁸ 8	3	⁴⁸ 4	1	5	6
²³⁶ 3	1	⁷¹⁸ 4	6	²⁴⁸ 2	5	7	²⁹ 9	⁸ 8
5	6	³⁸ 8	²⁹ 9	7	1	3	4	²⁹ 2
³⁶ 4	2	1	7	5	³⁴ 3	6	8	³⁷ 9
7	4 5	6	1	9	³⁴⁸ 8	⁴⁵ 4	²⁸ 2	⁶³⁹ 3
9	8 3	³¹ 3	²⁵⁸ 2	4	³⁴⁸³ 6	⁴⁵ 5	7	1

3	2	8	6	5	9	1	7	4
6	7	9	2	4	1	5	8	3
1	4	5	8	3	7	6	2	9
9	5	4	1	8	6	7	3	2
7	1	3	4	9	2	8	5	6
2	8	6	5	7	3	4	9	1
4	3	7	9	6	5	2	1	8
8	9	1	7	2	4	3	6	5
5	6	2	3	1	8	9	4	7

8	4	9	5	3	1	6	2	7
1	5	3	2	6	7	9	8	4
2	6	7	4	8	9	3	1	5
6	3	4	1	5	8	7	9	2
9	7	2	6	4	3	1	5	8
5	1	8	9	7	2	4	3	6
3	2	6	8	9	4	5	7	1
7	8	5	3	1	6	2	4	9
4	9	1	7	2	5	8	6	3

2

✓✓✓
27

✓✓✓

✓✓✓

4	1	7	2	5	9	6	3	8
9	3	5	8	7	6	2	4	1
8	6	2	4	1	3	9	7	5
5	4	1	7	2	8	3	6	9
6	7	8	9	3	5	1	2	4
3	2	9	6	4	1	8	5	7
2	9	4	3	8	7	5	1	6
1	8	3	5	6	4	7	9	2
7	5	6	1	9	2	4	8	3

✓
3 5

9	6	1	4	2	5	7	8	3
3	5	4	9	7	8	2	1	6
2	8	7	1	6	3	9	4	5
1	3	9	7	8	2	6	5	4
7	4	6	3	5	1	8	2	9
5	2	8	6	9	4	1	3	7
8	9	5	2	3	7	4	6	1
6	1	2	5	4	9	3	7	8
4	7	3	8	1	6	5	9	2



6	5	3	4	2	9	1	8	7
9	1	7	5	3	8	2	4	6
4	2	8	7	1	6	3	5	9
2	7	4	6	8	3	5	9	1
3	9	6	2	5	1	8	7	4
5	8	1	9	4	7	6	2	3
1	3	2	8	9	4	7	6	5
7	4	5	3	6	2	9	1	8
8	6	9	1	7	5	4	3	2

✓ ✓
2 7

, 11

4

2	7	9	8	5	3	1	6	5
3	6	1	9	4	7	5	2	8
8	4	5	2	6	1	9	7	3
1	9	7	3	8	5	2	4	6
4	5	8	6	7	2	3	1	9
6	2	3	1	9	4	8	5	7
9	1	4 _x	7	2	8	6 _x	3	5
5	3	6	4 _x	1	9	7	8	2
7	8	2	5	3	6	4 _x	9	1

✓✓
19



9	7	4	1	6	5	8	2	3
5	6	2	3	8	4	9	7	1
1	3	8	2	7	9	6	4	5
7	4	1	8	3	6	2	5	9
3	8	9	5	2	1	4	6	7
2	5	6	9	4	7	3	1	8
6	9	3	7	1	2	5	8	4
8	2	7	4	5	3	1	9	6
4	1	5	6	9	8	7	3	2

✓
5 7

4	3	7	9	8	5	1	2	6
2	9	1	8 6	7	3	4	5	8
5	6	8	1	4	2	9	7	3
3	7	5	8	2	9	6	4	1
6	8	4	5	1	7	2	3	9
1	2	9	3	6	4	5	8	7
9	1	3	2	5	8	7	6	4
7	5	6	4	3	1	8	9	2
8	4	2	7	9	6	3	1	5

✓
1 2 3 4 5 6 7 8 9 ✓
—

8 6	2	5 3	4	1	9 7	7		
4	5	3	9	1	7	2	6	8
7	1	9 6	8	2 5	4	3		
5	7	4	1	2	6	3	8	9
3	2	8 4	5	9	6 7	1		
6 9	1	3	7 8	4 5	2			
1	8 6	2	9	5	7	3	4	
2 4	7 8	6 3	9	1	5			
9 3	5	7	4 1	8	2	6		

1 2 3 4 5 6 7 8



7	3	5	6	4	2	9	8	1
2	9	8	3	5	1	4	7	6
1	4	6	8	7	9	5	3	2
4	2	9	7	1	6	8	5	3
6	7	3	2	8	5	1	4	9
8	5	1	9	3	4	2	6	7
5	6	7	1	2	8	3	9	4
9	1	4	5	6	3	7	2	8
3	8	2	4	9	7	6	1	5

1 2 3 4 5 6 7 8 9 ✓

6	3	2	8 ⁵	7	⁵	8	9	4
8	5	1	8	2	9	7	3	6 ^x
9	7	8	3	6	4	5	1 ^x	2
7	9	6	8	4	2 ^x	3	5	1
2	8	3		5		4	6	9
1	4	5	9	3	6	2	7	8
5	^x	9	^x	8	^x ^x	1	2	3
8	1	7	2	9	3	6	4	5
3				1	^x	9	8	7
6	3	2	5	7	1	8	9	4
4	5	1	8	2	9	7	3	6
9	7	8	3	6	4	5	1	2
7	9	6	1 ^x	4	2	3	5	8 ^x
2	1	3	7 ^x	5	8 ^x	4	6	9
8	4	5	9	3	6	2	7	1 ^x
5	6	9	4	8	7	1	2	3
1	8	7	2	9	3	6	4	5
3	2	4	6	1	5	9	8	7

3	5	8	6	9	7	2	4	1
1	9	2	4	8	5	6	7	3
6	7	4	2	1	3	5	9	8
5	3	9	8	7	1	4	6	2
8	6	7	3	4	2	9	1	5
4	2	1	9	5	6	8	3	7
2	1	3	5	6	9	7	8	4
7	4	6	1	2	8	3	5	9
9	8	5	7	3	4	1	2	6

✓
1 2 3 4 5 6 7 8 9

1 4 5 6 7 8

2	4	5	8	9	1	6	7	3
7	6	1	5	3	4	8	9	2
8	9	3	6	2	7	1	5	4
4	3	8	7	5	2	9	1	6
6	5	9	4	1	8	3	2	7
1	7	2	9	6	3	5	4	8
5	8	4	1	7	6	2	3	9
3	1	6	2	4	9	7	8	5
9	2	7	3	8	5	4	6	1

✓
1 2 3 4

4	7	1	6	8	9	5	2	3
3	9	6	2	7	5	8	1	4
8	2	5	4	1	3	7	9	6
1	3	2	7	9	4	6	8	5
7	5	8	3	2	6	9	4	1
6	4	9	1	5	8	3	7	2
2	1	3	9	6	7	4	5	8
9	8	4	5	3	1	2	6	7
5	6	7	8	4	2	1	3	9

✓
1 2 3 4 5 6 7 8 9

1

124

8	6	7	1	5	4	3	9	2
5	9	1	2	6	3	7	4	8
2	4	3	9	7	8	1	5	6
1	8	4	6	2	7	5	3	9
6	3	5	4	8	9	2	7	1
7	2	9	3	1	5	8	6	4
9	1	2	7	3	6	4	8	5
4	7	8	5	9	1	6	2	3
3	5	6	8	4	2	9	1	7

2

1 2 3 4 5 6 7 8 9

¹³ 3	4	7	² 2	¹³ 1	9	¹³ 8	6	5
5	⁶⁸ 8	²⁶⁸ 6	4	⁹³⁶ 3	7	1	2	9
2	¹⁶ 1	9	5	¹⁶ 6	8	7	⁴⁷ 4	13
³⁸ 8	9	2	6	5	4	³⁸ 3	1	7
¹⁷ 1	7	¹⁵ 5	² 3	9	2	⁴⁷ 4	⁹³⁸ 8	6
6	3	⁴⁵ 4	7	8	1	9	¹⁵ 5	2
⁴ 4	5	8	9	²⁴ 2	3	²⁶ 6	7	1
9	¹⁶¹⁸ 6	¹⁶ 1	8	⁹²¹ 7	5	²⁶ 2	3	4
7	2	³² 3	1	⁴⁷ 4	6	5	9	8

✓✓✓✓✓✓

1 2 3 4 5 6 7 8 9

1.78

3.6

9	8	4	1	6	2	7	3	5
7	5	6	8	9	3	1	2	4
1	2	3	5	7	4	8	6	9
3	7	8	2	5	1	4	9	6
4	6	5	3	8	9	2	1	7
2	9	1	7	4	6	3	5	8
6	4	2	9	3	7	5	8	1
5	1	9	4	2	8	6	7	3
8	3	7	6	1	5	9	4	2



✓✓ ✓✓✓✓
1 2 3 4 5 6 7 8 9

3	2	8	4	6	1	9	5	7
5	1	9	3	7	2	6	4	8
4	6	7	9	8	5	3	2	1
2	9	1	7	3	8	5	6	4
8	4	5	2	1	6	7	3	9
7	3	6	5	9	4	1	8	2
9	5	2	6	4	7	8	1	3
6	8	3	1	2	9	4	7	5
1	7	4	8	5	3	2	9	6

5-10-11

✓✓✓✓✓✓✓
 1 2 3 4 5 6 7 8 9

~~10~~

6	5	3	4	1	2	7	8	9	135
9	7	8	5	3	6	2	1	4	
2	4	1	9	8	7	3	6	5	
8	9	4	2	7	3	1	5	6	
3	1	7	6	9	5	4	2	8	
5	2	6	8	4	1	9	3	7	
7	8	2	3	6	4	5	9	1	
4	3	9	1	5	8	6	7	2	
1	6	5	7	2	9	8	4	3	→ 348

348

9	5	6	8	2	4	7	1	3
2	1	8	9	3	7	4	6	5
4	7	3	5	1	6	8	2	9
7	8	4	1	5	2	3	9	6
5	3	9	6	7	8	1	4	2
6	2	1	3	4	9	5	7	8
3	6	2	4	8	1	9	5	7
8	4	7	2	9	5	6	3	1
1	9	5	7	6	3	2	8	4

✓ ✓ ✓ 93
 1 2 3 4 5 6 7 8
 ✓
 6 9 18 4
~~3567~~
2357
 27 35

3	6	2	4	9	8	5	1	7
4	1	9	7	5	3	2	6	8
5	8	7	6	2	1	4	3	9
8	2	6	5	4	7	1	9	3
9	5	3	1	6	2	7	8	4
7	4	1	8	3	9	6	2	5
1	3	8	2	7	5	9	4	6
2	7	4	9	8	6	3	5	1
6	9	5	3	1	4	8	7	2

6	8	2	1	4	9	7	3	5
1	9	5	2	3	7	4	6	8
3	7	4	5	8	6	9	1	2
7	2	1	4	5	8	6	9	3
4	3	8	6	9	2	5	7	1
5	6	9	7	1	3	2	8	4
2	1	6	8	7	5	3	4	9
8	5	3	9	6	4	1	2	7
9	4	7	3	2	1	8	5	6

✓
6 7 8 9 ✓

5	2	9	7	1	4	3	6	8
4	3	1	6	9	8	7	5	2
6	8	7	3	5	2	1	9	4
9	1	8	4	7	6	2	3	5
3	6	4	8	2	5	9	7	1
7	5	2	1	3	9	4	8	6
8	7	5	9	4	1	6	2	3
2	4	3	5	6	7	8	1	9
1	9	6	2	8	3	5	4	7

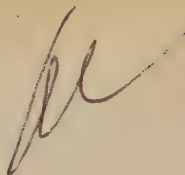
1 2 3 4 5 6 7 8 9

6	1	9	7	5	8	2	4	3
5	4	8	6	3	2	7	1	9
3	2	7	9	4	1	5	8	6
4	8	1	2	9	3	6	5	7
7	6	3	5	8	4	1	9	2
2	9	5	1	6	7	4	3	8
1	3	6	4	2	9	8	7	5
9	7	2	8	1	5	3	6	4
8	5	4	3	7	6	9	2	1

1 2 3 4 5 6 7 8 9

3	2	6	1	4	5	9	8	7
8	4	7	9	6	3	2	1	5
1	5	9	2	8	7	4	6	3
6	1	4	8	5	9	3	7	2
2	8	3	6	7	4	1	5	9
9	7	5	3	2	1	8	4	6
4	3	2	7	1	6	5	9	8
7	9	1	5	3	8	6	2	4
5	6	8	4	9	2	7	3	1

6	3	1	8	9	2	7	5	4
5	8	7	1	4	3	9	6	2
2	4	9	7	6	5	1	3	8
3	6	5	9	1	8	2	4	7
7	1	8	5	2	4	3	9	6
9	2	4	3	7	6	8	1	5
1	7	6	2	5	9	4	8	3
4	9	3	6	8	7	5	2	1
8	5	2	4	3	1	6	7	9



6	9	1	5	7	4	3	2	8
4	7	2	8	6	3	5	1	9
8	5	3	1	2	9	7	6	4
7	1	8	2	4	5	6	9	3
3	2	6	9	1	8	4	7	5
5	4	9	6	3	7	2	8	1
9	3	7	4	8	2	1	5	6
1	8	4	7	5	6	9	3	2
2	6	5	3	9	1	8	4	7

9	4	6	7	5	8	1	2	3
1	3	8	6	4	2	5	9	7
5	2	7	9	1	3	4	8	6
4	1	9	2	8	6	7	3	5
6	7	2	3	9	5	8	4	1
3	8	5	1	7	4	9	6	2
8	6	1	4	3	7	2	5	9
2	9	4	5	6	1	3	7	8
7	5	3	8	2	9	6	1	4

7	1	8	6	2	4	9	3	5
9	2	4	3	1	5	6	7	8
5	3	6	7	9	8	4	1	2
6	5	3	2	4	7	8	9	1
1	7	2	8	6	9	5	4	3
8	4	9	5	3	1	2	6	7
4	8	5	9	7	3	1	2	6
2	9	7	1	8	6	3	5	4
3	6	1	4	5	2	7	8	9

ramj

4	2	8	3	5	6	7	1	9
5	9	1	8	4	7	3	6	2
6	3	7	9	2	1	4	5	8
3	8	6	2	7	5	9	4	1
9	7	4	6	1	8	5	2	3
1	5	2	4	3	9	8	7	6
8	4	9	5	6	2	1	3	7
2	1	5	7	9	3	6	8	4
7	6	3	1	8	4	2	9	5

1 2 3 4 5 6 7 8 9

3	2	6	7	1	9	8	4	5
8	7	5	3	6	4	1	9	2
9	1	4	8	5	2	6	7	3
4	9	8	6	3	5	2	1	7
7	6	3	2	9	1	5	8	4
1	5	2	4	8	7	9	3	6
5	4	1	9	7	6	3	2	8
6	8	7	1	2	3	4	5	9
2	3	9	5	4	8	7	6	1

1 2 3 4 5 6 7 8 9

5	1	3	4	9	7	6	8	2
7	9	8	1	2	6	4	3	5
4	2	6	5	8	3	7	9	1
8	5	2	9	6	1	3	4	7
9	7	1	3	5	4	2	6	8
6	3	4	2	7	8	5	1	9
3	4	9	7	1	5	8	2	6
2	6	5	8	4	9	1	7	3
1	8	7	6	3	2	9	5	4

123456789

8	3	6	5	1	9	2	7	4
1	5	2	8	7	4	9	6	3
4	7	9	3	2	6	5	8	1*
7	8	4	6	9	3	1	5	2
5	2	1	7	4	8	3	9	6
6	9	3	1	5	2	7	4	8
3	4	7	2	8	5	6	1	9
9	6	5	4	3	1	8	2*	7
2	1	8	9	6	7	4	3	5

[2] 3 4 5 6 7 8 9

- 300. 100

2	3	7	9	4	5	6	1	8
6	1	5	7	8	3	4	9	2
9	4	8	6	2	1	7	3	5
4	7	6	2	9	8	3	5	1
1	9	2	3	5	7	8	4	6
5	8	3	4	1	6	2	7	9
8	6	4	1	7	9	5	2	3
3	2	1	5	6	4	9	8	7
7	5	9	8	3	2	1	6	4

1 2 3 4 5 6 7 8 9

8	6	2	5	4	7	9	1	3
9	1	3	8	6	2	7	4	5
7	4	5	3	9	1	6	8	2
5	2	8	9	7	4	1	3	6
1	7	6	2	8	3	5	9	4
4	3	9	1	5	6	8	2	7
3	8	1	7	2	5	4	6	9
2	5	4	6	1	9	3	7	8
6	9	7	4	3	8	2	5	1

1 2 3 4 5 6 7 8 9

3				..				7
9		1		.	4	3		.
		7		8			5	.
		6	4	5	..			3
			2	3	9	..		.
4			.	6	..	5		..
	2		.	9	.	4		5
.		4	8	.	5	2	.	6
5			x	.	4	2		1

x	1	5	9	8	7			
8	6	7		4				1
9							7	8
7			1	9		8		
	5						3	
		8		2	4			7
4	8							9
5							4	
			4	5	9	2	8	

			5	6				
		3		8	7			
		8			9	6	1	
	6	7						3
5	2						6	7
8						1	4	
	9	1	4			8		
			7	9		4		
				2	8			

		7						
	3				4		6	
		6		3	2	5		8
	5	1		2				
		4	1		8	2		
				7		9	3	
4		9	3	1		6		
	2		5				9	
						8		

	3	4		8	2			1
9				4			8	
2								
			1	3				5
6	5		9		4		7	3
1				6	5			
								6
	6			7				9
3			6	5		2	4	

8	4			1		5		7
5		6			3		2	
	1							8
			9				5	
2				7				4
	6				8			
6							9	
	5		3			4		2
1		8		9			7	5

7			2				4	9
6		1	9			7		
	9						3	
				9			5	4
			8		5			
8	2			1				
	3						7	
		7			4	2		3
4	6				7			5

	4				3	5		
	5		6				2	
		9		2		8	3	
		1	9	4				
				8	2	6		
	6	8		1		7		
	7				9		5	
		4	7				1	

1 2 3 4 5 6 7 8 9

.	.	5	x	.	.	3	8	.	.
3	.	8	.	7	.	4	.	.	x
7	4	1	5	8	9	2	6	3	.
4	x	9	x	.	.	7	.	.	.
.	7	6	.	.	.	3	2	x	x
.	.	3	.	.	.	6	x	1	.
.	3	7	1	9	.	5	.	2	.
.	x	2	.	5	.	1	.	.	.
.	.	4	8	.	.	9	.	.	.

5								1
		4	6		1	3		
	1		3		5		8	
		3		5		1		
9	7						6	4
		1		6		2		
	9		2		7		1	
		8	5		6	9		
3								6

1		4						7
	3		1		8			
5				7				
	2			5		9		
4	5						7	2
		6		3			5	
				2				4
			6		3		9	
2						1		8

					3		7	1
	3	6			7			8
	7		8			5		
		9					8	7
				9				
5	8					4		
		2			6		3	
7			4			9	1	
4	1		3					

			7		5	4		
		3	6					
1		7	8			5	3	
9						1	4	2
7	3	1						5
	8	6			7	2		1
					6	9		
		9	2		8			

Super Fiendish

.

✓✓✓
1 2 3 4 5 6 7 8 9

9	7	5	4	1	2	8	3	6
1	8	3	6	5	7	2	9	4
6	4	2	9	8	3	7	5	1
8	9	4	1	6	5	3	2	7
2	5	7	8	3	4	1	6	9
3	1	6	2	7	9	4	8	5
4	3	8	7	9	6	5	1	2
5	2	9	3	4	1	6	7	8
7	6	1	5	2	8	9	4	3

~~1256~~

1 2 3 4 5 6 7 8 9 ✓

~~1256~~ 1256

9	7	6	1	2	4	5	1	2	4	3	8
3	1	5	8	6	4	7	7	1	5	4	9
4	2	6	5	2	3	9	8	6	5	1	7
5	5	6	1	1	6	4	7	4	9	8	2
8	9	9	7	5	5	2	3	3	1	4	4
2	4	3	8	1	6	7	5	9			
1	1	3	9	2	8	4	2	6	5	5	
6	8	2	4	1	1	3	3	5	9	7	3
7	5	2	4	9	6	2	3	2	8	1	

~~1256 1256 1256 1256~~

* numbers on left are the answers.

2

7	3	7	2	9	8	5	6	4	1
	9	4	6	7	3	1	8	5	2
8	8	1	5	2	6	4	7	9	3
	2	6	3	4	9	7	1	8	5
	1	9	8	5	2	3	4	6	7
	4	5	7	8	1	6	2	3	9
8	7	2	4	6	5	9	3	1	8
	6	3	9	1	7	8	5	2	4
	5	8	1	3	4	2	9	7	6

30000000 m/s

$\checkmark$ $\checkmark$ $\checkmark$
 1 2 3 4 5 6 7 8 9

8
 1308

4	5	8	6	8	7	9	1	2
7	6	8	1	9	2	4	3	5
1	9	2	5	4	3	7	6	8
9	7	5	8	3	4	6	2	1
2	4	1	9	6	5	3	8	7
3	8	6	2	7	1	5	9	4
8	3	4	7	2	9	1	5	6
6	1	9	4	5	8	2	7	3
5	2	7	3	1	6	8	4	9

2577

125

1 2 3 4 5 6 7 8 9

2	5	9	8	4	1	7	6
8	2	1	7	4	3	9	
4	1	7	3	9	6	5	8
5	2	6	9	4	8	7	1
3	7	4	1	2	3	8	6
9	7	5	4	3	1	6	2
1	4	2	8	6	9	3	5
6	3	7	5	2	9	1	8

3 5 7

✓ ✓✓
1 2 3 4 5 6 7 8 9

245

459

7	8	1	5	3	4	9	2	6
4	5	6	7	9	2	1	3	8
3	9	2	1	6	8	5	7	4
6	2	7	4	8	5	3	1	9
5	1	3	9	2	6	8	4	7
8	4	9	3	7	1	2	6	5
9	6	8	2	4	3	7	5	1
1	3	4	8	5	7	6	9	2
2	7	5	6	1	9	4	8	3
7	8	1	5					
x	5	6	7			1	x	8
x	x	x	1	6	8		7	
6	2	7	4	8	5	3	1	9
5	1	3	9	2	6	8		7
8	4	9	3	7	1	2		
9	6	8	2	4	3	7		1
1			8	5	7	6		
x	7		6	1	9		8	

1 2 3 4 5 6 7 8 9

9	x		7	1	6	2		
	5		2	9	4		6	
2			8	3	5		7	9
	2		9	8		x	4	x
3	1	9	5	4	7	6	2	8
x	7	x	6	2			9	
1			3		9			2
	9		4		2		1	
			2	1	7	8	9	

	9	4	2			5		
3					4		9	
				5		4		1
	7							
4			8	7	5			6
							8	
7	4	6		3				
..	2	..	5	4	6	.	.	3
5	3	1	7	..	2	6	4	..

9	4		2		8			1
		1	4					
		3	9	1	6	7		4
4			1	6	5	8		2
		8		4	2	6		
6		2	8		9			5
		4		9	1	5		
			5	8				
1			6	2	4			8

	5				8		9	
1	7			6				
8				9		5		
		1	3	2				
		5				7		
				4	9	1		
		8		7				2
				5			3	4
	4		1				7	

				9	7	3	5	6
					6			2
5		7			3			
	8	4						
6								5
						4	1	
			1			8		9
1			8					
8	4	9	6	3				

		3	2	6			4	
9			7			1		
	8			9				7
							6	1
7		9				8		5
6	5							
4				8			9	
		2			4			8
	6			1	3	2		

	1	9		6	4			
	8	3						
5		2	9		3			
	9	4			2			
		8				3		
			8			6	4	
			5		9	4		2
						9	1	
			1	4		8	3	

2				8				4
		4	2		9			
		7				3	2	
	8						5	
1				6				2
	7						3	
	1	5				8		
			1		8	7		
6				5				3

					5	1		3
			3	8	1	6		
	1			9				8
	2							9
9			5	3	6			2
8							4	
2				6			7	
		9	7	5	8			
6		5	2					

		9		7	8	3		1
3		4		1			7	
6					9			
2								6
	9						8	
1								3
			9					5
	5			2		6		7
8		6	3	5		2		

	2	5			8	6	1	
		1				5		2
7	5			9			3	
3			1	8	7			5
	8			5			4	9
9		7				3		
	6	8	4			9	2	

7	9		6	1				4
		1			2		6	
5		2	9					7
			2					
8		9				7		2
					4			
1					6	8		9
	3		4			6		
6				7	9		2	3

1			4			5		2
	9		1	3		4		
	7		9	1			3	
		4		8		9		
	1			6	3		2	
		1		7	6		4	
5		6			9			8

8				7			6	
	4	2	9					7
	3					8		
	6			9	5			
5			3		7			8
			8	1			9	
		6					4	
3					1	9	8	
	5			3				2

	8			9			1	
4	5			8				
							5	7
				5	7			9
	9		3		8		7	
7			6	4				
2	3							
				6			8	2
	1			3			9	

		9			4	2	7	
8								
4				1	7			3
2		6		9				
		7	3	5	8	4		
				7		1		9
9			5	4				7
								2
	5	1	7			9		

		2		9	1	6		
			4				1	
			7				9	3
3		5	9			2		
6								5
		1			5	3		8
4	8				9			
	5				7			
		9	1	4		8		

	6	2	3			7		8
	9		2			3	6	
	1				3	4		
		7	6		5	9		
		4	1				2	
	5	1			6		4	
2		9			7	5	1	

9								6
		2	1		7	4		
	8		6				2	
	9			3		8	5	
			4		5			
	3	4		6			1	
	1				4		6	
		9	5		1	2		
8								7

9								4
	2				5		7	
		7	2	1		6		
	5		8		7	9		
		3				7		
		8	1		9		4	
		6		8	1	5		
	7		6				2	
8								6

			2	4	5		7	
4								3
2					7		5	9
8		6	7		9			
			3		8	9		7
1	2		5					8
7								5
	8		4	9	2			

			6		9			
	1		8		7		6	
		9		5		4		
7		6				1		9
9	4						2	5
2		1				6		3
		2		8		5		
	3		2		5		9	
			9		3			

4							2	6
				3			5	
		9	2			7		
	2		7		1			
		5		2		6		
			4		5		3	
		6			7	8		
	7			1				
1	4							9

6				5		9		1
		1					5	
5					8			4
		6		8			2	
			1		3			
	5			2		3		
4			8					6
	9					5		
2		7		4				8

Solutions

1

8	2	1	3	7	4	5	6	9
3	4	7	5	9	6	1	2	8
5	6	9	8	2	1	7	3	4
1	3	8	4	5	9	2	7	6
9	5	6	7	8	2	3	4	1
2	7	4	1	6	3	9	8	5
6	8	2	9	1	7	4	5	3
4	1	5	2	3	8	6	9	7
7	9	3	6	4	5	8	1	2

2

3	9	7	5	4	2	1	6	8
5	1	2	6	7	8	4	3	9
4	6	8	9	3	1	7	2	5
1	7	5	8	6	9	2	4	3
6	8	3	2	5	4	9	1	7
2	4	9	7	1	3	5	8	6
7	5	4	1	8	6	3	9	2
8	2	1	3	9	5	6	7	4
9	3	6	4	2	7	8	5	1

3

2	1	3	6	9	7	4	5	8
4	8	6	5	2	1	9	7	3
5	9	7	8	3	4	2	1	6
3	2	1	9	4	5	8	6	7
7	5	9	3	8	6	1	2	4
6	4	8	1	7	2	3	9	5
8	3	5	7	1	9	6	4	2
9	7	4	2	6	3	5	8	1
1	6	2	4	5	8	7	3	9

4

5	4	3	2	8	7	9	6	1
6	2	9	1	3	4	5	7	8
7	1	8	5	9	6	2	4	3
2	6	7	3	1	9	8	5	4
4	8	1	6	7	5	3	9	2
3	9	5	8	4	2	6	1	7
9	3	4	7	5	8	1	2	6
8	5	6	4	2	1	7	3	9
1	7	2	9	6	3	4	8	5

5

3	4	7	6	8	1	2	5	9
6	2	8	5	9	3	1	4	7
9	1	5	4	7	2	6	3	8
1	3	9	2	6	4	8	7	5
8	7	6	1	5	9	3	2	4
2	5	4	8	3	7	9	6	1
5	6	3	9	4	8	7	1	2
7	9	1	3	2	5	4	8	6
4	8	2	7	1	6	5	9	3

6

7	3	5	8	1	6	2	9	4
4	1	9	2	5	7	3	8	6
6	8	2	3	9	4	5	1	7
1	4	7	5	3	8	6	2	9
5	9	6	4	2	1	8	7	3
8	2	3	6	7	9	1	4	5
3	5	4	7	8	2	9	6	1
2	7	1	9	6	5	4	3	8
9	6	8	1	4	3	7	5	2

7

7	5	2	8	4	9	3	1	6
3	1	8	6	7	5	4	2	9
6	4	9	3	1	2	7	8	5
1	9	4	2	3	7	5	6	8
2	7	5	1	6	8	9	3	4
8	3	6	5	9	4	2	7	1
9	8	3	4	2	1	6	5	7
5	2	7	9	8	6	1	4	3
4	6	1	7	5	3	8	9	2

8

5	2	8	9	7	3	4	6	1
7	1	6	8	2	4	3	9	5
3	4	9	5	1	6	7	8	2
6	9	7	1	8	5	2	3	4
8	3	1	2	4	9	5	7	6
2	5	4	6	3	7	9	1	8
4	8	3	7	6	2	1	5	9
9	6	2	3	5	1	8	4	7
1	7	5	4	9	8	6	2	3

9

3	5	6	4	1	7	8	2	9
2	8	1	3	6	9	7	4	5
4	7	9	2	5	8	3	1	6
5	9	4	8	7	1	6	3	2
1	6	3	5	2	4	9	7	8
8	2	7	6	9	3	1	5	4
6	1	2	9	3	5	4	8	7
9	3	8	7	4	2	5	6	1
7	4	5	1	8	6	2	9	3

10

1	5	9	4	2	6	3	7	8
8	4	6	3	1	7	2	9	5
2	3	7	8	5	9	6	4	1
6	8	5	7	4	3	1	2	9
4	2	1	9	6	5	8	3	7
7	9	3	2	8	1	4	5	6
5	7	8	6	3	4	9	1	2
9	6	4	1	7	2	5	8	3
3	1	2	5	9	8	7	6	4

11

2	5	3	1	9	4	6	7	8
7	4	1	8	3	6	9	2	5
9	8	6	2	5	7	4	1	3
6	7	2	4	8	9	3	5	1
4	3	5	6	7	1	8	9	2
1	9	8	5	2	3	7	6	4
8	6	9	3	1	2	5	4	7
3	1	7	9	4	5	2	8	6
5	2	4	7	6	8	1	3	9

12

5	3	2	6	9	1	4	7	8
1	7	9	2	8	4	6	5	3
8	4	6	5	7	3	2	9	1
3	8	5	4	1	7	9	6	2
2	1	4	3	6	9	7	8	5
9	6	7	8	5	2	3	1	4
6	2	8	9	3	5	1	4	7
4	5	1	7	2	6	8	3	9
7	9	3	1	4	8	5	2	6

13

9	8	3	7	4	1	5	6	2
7	2	5	8	6	9	3	1	4
1	4	6	2	3	5	8	9	7
5	1	8	3	7	2	9	4	6
4	6	7	1	9	8	2	3	5
2	3	9	6	5	4	7	8	1
3	5	4	9	1	7	6	2	8
6	7	2	4	8	3	1	5	9
8	9	1	5	2	6	4	7	3

14

4	1	5	3	7	9	2	8	6
6	2	3	5	1	8	7	4	9
8	9	7	6	4	2	5	3	1
7	4	2	1	3	6	9	5	8
5	3	9	4	8	7	6	1	2
1	8	6	9	2	5	3	7	4
9	6	1	7	5	4	8	2	3
2	5	4	8	9	3	1	6	7
3	7	8	2	6	1	4	9	5

15

6	7	3	4	5	9	1	8	2
1	4	9	6	2	8	5	7	3
2	8	5	3	1	7	9	4	6
5	1	2	9	7	6	4	3	8
9	3	8	2	4	5	7	6	1
4	6	7	8	3	1	2	9	5
3	5	4	7	6	2	8	1	9
8	2	6	1	9	4	3	5	7
7	9	1	5	8	3	6	2	4

16

4	5	2	6	3	1	7	8	9
7	3	1	2	9	8	5	6	4
9	8	6	5	7	4	1	2	3
1	4	3	8	2	9	6	7	5
2	6	5	7	1	3	4	9	8
8	9	7	4	5	6	3	1	2
3	7	9	1	8	5	2	4	6
6	2	8	3	4	7	9	5	1
5	1	4	9	6	2	8	3	7

17

3	1	9	4	6	2	7	5	8
2	8	5	9	3	7	4	1	6
4	7	6	5	1	8	3	2	9
9	6	1	2	7	5	8	4	3
8	2	4	1	9	3	6	7	5
5	3	7	6	8	4	1	9	2
6	5	8	7	4	9	2	3	1
1	4	2	3	5	6	9	8	7
7	9	3	8	2	1	5	6	4

18

8	4	6	7	1	9	3	2	5
7	5	1	4	3	2	8	6	9
2	9	3	5	8	6	7	1	4
6	3	7	1	5	8	4	9	2
5	1	2	3	9	4	6	8	7
9	8	4	6	2	7	5	3	1
4	6	8	2	7	1	9	5	3
1	7	5	9	6	3	2	4	8
3	2	9	8	4	5	1	7	6

19

5	7	2	9	8	6	1	3	4
3	6	1	7	2	4	9	8	5
9	4	8	5	1	3	2	6	7
2	5	3	4	9	1	8	7	6
4	9	7	6	5	8	3	1	2
8	1	6	2	3	7	5	4	9
7	3	9	8	6	5	4	2	1
1	2	4	3	7	9	6	5	8
6	8	5	1	4	2	7	9	3

20

7	1	2	6	8	3	9	5	4
6	9	4	1	2	5	3	7	8
8	3	5	9	4	7	2	1	6
5	8	6	7	3	9	4	2	1
4	2	1	5	6	8	7	9	3
9	7	3	2	1	4	8	6	5
2	4	7	8	5	1	6	3	9
3	5	9	4	7	6	1	8	2
1	6	8	3	9	2	5	4	7

21

6	5	1	2	8	9	3	7	4
2	8	4	1	3	7	9	6	5
3	7	9	4	5	6	8	2	1
4	3	5	9	2	1	6	8	7
9	2	7	6	4	8	5	1	3
1	6	8	3	7	5	4	9	2
7	4	3	8	9	2	1	5	6
5	9	6	7	1	4	2	3	8
8	1	2	5	6	3	7	4	9

22

6	2	7	8	4	3	1	9	5
8	9	1	7	6	5	4	3	2
5	3	4	9	2	1	6	8	7
7	4	9	5	3	6	2	1	8
3	5	8	1	9	2	7	6	4
2	1	6	4	8	7	9	5	3
9	6	3	2	7	8	5	4	1
4	7	5	3	1	9	8	2	6
1	8	2	6	5	4	3	7	9

23

3	2	7	9	8	4	1	6	5
8	9	1	2	5	6	4	3	7
6	5	4	1	3	7	8	9	2
5	7	9	6	4	8	2	1	3
4	8	2	5	1	3	9	7	6
1	3	6	7	2	9	5	8	4
7	4	8	3	9	2	6	5	1
2	6	5	8	7	1	3	4	9
9	1	3	4	6	5	7	2	8

24

1	8	6	5	9	2	3	7	4
4	7	2	1	3	8	5	9	6
5	9	3	4	6	7	8	1	2
6	1	8	7	4	9	2	3	5
3	4	5	2	1	6	7	8	9
9	2	7	3	8	5	4	6	1
7	5	9	6	2	3	1	4	8
8	3	4	9	5	1	6	2	7
2	6	1	8	7	4	9	5	3

25

8	6	9	2	7	3	4	1	5
5	1	3	9	8	4	2	7	6
4	7	2	6	1	5	3	8	9
6	5	4	1	3	8	7	9	2
3	2	7	4	6	9	8	5	1
9	8	1	5	2	7	6	3	4
2	3	5	7	4	1	9	6	8
1	4	8	3	9	6	5	2	7
7	9	6	8	5	2	1	4	3

26

5	3	7	6	2	9	8	1	4
2	8	9	5	4	1	6	7	3
1	4	6	8	7	3	2	9	5
9	5	2	3	1	7	4	8	6
8	7	4	2	5	6	1	3	9
6	1	3	9	8	4	7	5	2
3	6	8	7	9	2	5	4	1
4	9	5	1	6	8	3	2	7
7	2	1	4	3	5	9	6	8

27

1	7	6	5	8	3	9	2	4
2	5	4	7	1	9	8	3	6
9	8	3	6	4	2	5	1	7
4	1	5	9	6	8	2	7	3
3	9	2	1	5	7	6	4	8
7	6	8	2	3	4	1	9	5
6	3	1	4	9	5	7	8	2
5	4	7	8	2	1	3	6	9
8	2	9	3	7	6	4	5	1

28

3	4	7	6	9	5	2	8	1
1	6	9	2	7	8	3	4	5
8	2	5	4	1	3	7	6	9
2	5	6	8	3	9	1	7	4
7	1	3	5	4	6	9	2	8
4	9	8	1	2	7	6	5	3
6	7	1	9	5	4	8	3	2
5	8	2	3	6	1	4	9	7
9	3	4	7	8	2	5	1	6

29

6	3	2	7	5	8	9	1	4
7	4	8	1	9	6	3	5	2
5	9	1	3	4	2	6	8	7
3	5	4	2	7	1	8	9	6
9	1	7	8	6	5	2	4	3
2	8	6	9	3	4	1	7	5
8	2	3	5	1	7	4	6	9
1	6	5	4	2	9	7	3	8
4	7	9	6	8	3	5	2	1

30

4	2	9	1	3	5	6	8	7
5	3	6	2	7	8	9	1	4
8	1	7	6	9	4	3	5	2
1	8	2	4	5	6	7	3	9
9	7	5	8	2	3	4	6	1
3	6	4	9	1	7	8	2	5
7	9	8	5	6	1	2	4	3
2	4	1	3	8	9	5	7	6
6	5	3	7	4	2	1	9	8

31

9	7	1	5	6	3	8	2	4
2	3	6	1	8	4	5	9	7
5	8	4	9	7	2	6	3	1
7	1	8	4	2	6	3	5	9
4	5	3	8	1	9	7	6	2
6	9	2	7	3	5	1	4	8
3	6	7	2	9	8	4	1	5
1	2	5	6	4	7	9	8	3
8	4	9	3	5	1	2	7	6

32

5	8	3	1	2	6	7	9	4
4	2	9	3	5	7	1	6	8
1	7	6	9	8	4	5	3	2
2	9	4	6	7	8	3	1	5
7	1	8	4	3	5	6	2	9
6	3	5	2	9	1	4	8	7
9	4	7	8	6	3	2	5	1
3	5	2	7	1	9	8	4	6
8	6	1	5	4	2	9	7	3

33

7	3	8	6	5	1	4	2	9
4	2	6	7	3	9	1	5	8
1	9	5	2	4	8	6	7	3
9	1	3	8	6	7	2	4	5
2	6	4	9	1	5	8	3	7
8	5	7	4	2	3	9	1	6
6	8	2	3	7	4	5	9	1
5	7	9	1	8	2	3	6	4
3	4	1	5	9	6	7	8	2

34

8	7	4	3	5	1	9	2	6
6	9	1	7	4	2	3	5	8
5	3	2	6	8	9	4	1	7
3	1	6	8	9	5	7	4	2
4	5	8	1	2	7	6	9	3
9	2	7	4	6	3	5	8	1
7	4	5	2	1	6	8	3	9
1	6	9	5	3	8	2	7	4
2	8	3	9	7	4	1	6	5

35

6	3	5	7	4	2	1	9	8
2	4	8	1	6	9	3	7	5
1	7	9	5	3	8	4	6	2
9	2	7	4	5	6	8	3	1
3	1	6	2	8	7	5	4	9
5	8	4	9	1	3	7	2	6
4	6	3	8	2	1	9	5	7
8	9	2	3	7	5	6	1	4
7	5	1	6	9	4	2	8	3

36

3	6	9	4	7	8	1	5	2
4	2	1	6	5	3	8	9	7
5	8	7	9	1	2	4	3	6
6	4	3	8	2	7	5	1	9
2	9	5	1	4	6	3	7	8
7	1	8	3	9	5	6	2	4
1	5	4	7	8	9	2	6	3
9	3	2	5	6	4	7	8	1
8	7	6	2	3	1	9	4	5

37

4	3	9	2	6	8	7	1	5
7	5	8	3	1	4	9	2	6
6	1	2	5	9	7	4	3	8
3	7	6	8	4	2	5	9	1
9	8	4	1	3	5	6	7	2
5	2	1	6	7	9	8	4	3
1	9	5	4	2	6	3	8	7
2	6	7	9	8	3	1	5	4
8	4	3	7	5	1	2	6	9

38

9	1	7	2	3	4	6	8	5
8	5	6	9	1	7	2	4	3
3	4	2	5	6	8	9	1	7
2	9	4	1	8	5	3	7	6
5	8	3	4	7	6	1	2	9
7	6	1	3	9	2	8	5	4
4	3	8	6	5	1	7	9	2
1	2	9	7	4	3	5	6	8
6	7	5	8	2	9	4	3	1

39

6	4	3	9	5	7	1	8	2
5	1	9	8	4	2	7	6	3
2	7	8	6	1	3	9	5	4
3	2	6	5	9	4	8	1	7
7	9	1	2	6	8	3	4	5
4	8	5	3	7	1	6	2	9
1	6	4	7	2	9	5	3	8
8	5	7	4	3	6	2	9	1
9	3	2	1	8	5	4	7	6

40

8	3	7	6	5	2	1	4	9
1	9	4	7	3	8	5	2	6
2	6	5	4	1	9	3	8	7
6	2	9	8	7	5	4	1	3
4	1	3	9	2	6	7	5	8
5	7	8	3	4	1	9	6	2
7	4	2	1	6	3	8	9	5
3	8	6	5	9	4	2	7	1
9	5	1	2	8	7	6	3	4

41

4	6	2	5	3	1	8	9	7
8	1	9	4	7	2	3	5	6
3	5	7	6	8	9	4	1	2
7	3	6	9	2	4	5	8	1
5	2	4	8	1	7	9	6	3
1	9	8	3	5	6	2	7	4
6	4	3	7	9	5	1	2	8
9	8	1	2	6	3	7	4	5
2	7	5	1	4	8	6	3	9

42

6	1	9	4	8	2	5	3	7
3	2	4	5	7	9	8	1	6
8	7	5	1	6	3	4	2	9
5	9	6	2	4	1	7	8	3
1	3	8	9	5	7	2	6	4
2	4	7	8	3	6	9	5	1
9	8	2	3	1	4	6	7	5
7	5	1	6	9	8	3	4	2
4	6	3	7	2	5	1	9	8

43

7	2	8	6	1	3	9	5	4
3	5	9	7	8	4	1	6	2
1	6	4	2	5	9	3	8	7
8	3	6	5	2	1	7	4	9
9	4	5	3	6	7	8	2	1
2	1	7	4	9	8	5	3	6
4	7	2	9	3	5	6	1	8
5	9	1	8	4	6	2	7	3
6	8	3	1	7	2	4	9	5

44

6	2	5	8	1	4	7	3	9
7	9	3	6	2	5	8	1	4
8	4	1	3	7	9	5	2	6
3	1	4	7	6	2	9	5	8
5	7	2	9	4	8	1	6	3
9	6	8	1	5	3	4	7	2
1	8	6	2	9	7	3	4	5
4	3	7	5	8	6	2	9	1
2	5	9	4	3	1	6	8	7

45

7	8	6	4	9	2	3	5	1
1	9	5	7	8	3	4	2	6
2	4	3	5	6	1	7	9	8
3	1	8	2	4	6	9	7	5
4	7	9	8	3	5	1	6	2
5	6	2	1	7	9	8	4	3
8	2	4	6	1	7	5	3	9
6	3	1	9	5	4	2	8	7
9	5	7	3	2	8	6	1	4

46

2	5	4	7	3	6	1	8	9
3	9	1	2	8	5	4	6	7
8	7	6	1	9	4	2	3	5
4	1	2	9	6	8	7	5	3
9	6	3	4	5	7	8	2	1
5	8	7	3	1	2	6	9	4
7	2	9	8	4	3	5	1	6
6	3	8	5	7	1	9	4	2
1	4	5	6	2	9	3	7	8

47

1	8	5	4	6	7	3	2	9
4	2	3	5	1	9	6	7	8
9	6	7	8	2	3	1	5	4
6	4	8	3	7	1	5	9	2
2	5	9	6	4	8	7	3	1
7	3	1	9	5	2	4	8	6
3	1	4	2	8	5	9	6	7
5	7	2	1	9	6	8	4	3
8	9	6	7	3	4	2	1	5

48

2	7	4	5	8	9	1	3	6
1	9	6	7	3	2	8	5	4
5	8	3	4	1	6	2	7	9
3	5	2	6	7	1	4	9	8
9	1	8	3	5	4	6	2	7
4	6	7	2	9	8	5	1	3
6	2	1	9	4	7	3	8	5
7	4	5	8	2	3	9	6	1
8	3	9	1	6	5	7	4	2

49

7	1	6	4	5	3	2	9	8
4	9	2	6	8	1	7	5	3
5	8	3	2	7	9	6	1	4
9	6	5	1	4	8	3	7	2
2	7	8	9	3	6	1	4	5
3	4	1	7	2	5	9	8	6
6	5	9	8	1	2	4	3	7
8	2	7	3	9	4	5	6	1
1	3	4	5	6	7	8	2	9

50

6	3	5	7	9	8	2	1	4
9	8	7	4	1	2	3	5	6
4	1	2	3	5	6	8	7	9
8	2	6	1	4	9	7	3	5
1	5	9	6	3	7	4	8	2
7	4	3	2	8	5	6	9	1
3	9	4	8	2	1	5	6	7
5	6	8	9	7	4	1	2	3
2	7	1	5	6	3	9	4	8

51

9	2	3	6	8	5	7	4	1
6	4	7	9	2	1	5	8	3
8	1	5	3	4	7	2	6	9
1	5	4	8	9	3	6	7	2
3	6	2	1	7	4	8	9	5
7	8	9	2	5	6	3	1	4
2	3	6	7	1	9	4	5	8
4	9	8	5	6	2	1	3	7
5	7	1	4	3	8	9	2	6

52

8	2	5	6	9	3	4	7	1
7	9	4	5	8	1	3	2	6
6	3	1	7	2	4	8	5	9
1	5	8	9	4	2	7	6	3
4	7	9	3	6	8	2	1	5
3	6	2	1	7	5	9	4	8
2	4	3	8	5	6	1	9	7
9	1	6	4	3	7	5	8	2
5	8	7	2	1	9	6	3	4

53

9	5	6	8	4	2	1	3	7
3	2	4	9	1	7	8	6	5
8	1	7	5	6	3	2	9	4
6	7	3	2	8	5	9	4	1
5	8	1	3	9	4	7	2	6
4	9	2	6	7	1	5	8	3
7	6	9	4	5	8	3	1	2
1	3	8	7	2	6	4	5	9
2	4	5	1	3	9	6	7	8

54

9	6	2	3	7	8	4	1	5
3	5	4	2	9	1	7	8	6
8	1	7	4	6	5	3	2	9
2	9	8	7	4	6	1	5	3
4	7	1	8	5	3	6	9	2
6	3	5	1	2	9	8	7	4
1	2	3	9	8	4	5	6	7
7	4	6	5	1	2	9	3	8
5	8	9	6	3	7	2	4	1

55

8	7	5	9	2	1	4	3	6
2	3	9	7	4	6	5	1	8
6	1	4	5	3	8	7	9	2
1	4	8	6	5	3	2	7	9
7	9	3	2	8	4	6	5	1
5	2	6	1	9	7	8	4	3
4	8	7	3	6	9	1	2	5
3	6	2	4	1	5	9	8	7
9	5	1	8	7	2	3	6	4

56

6	2	1	4	9	8	5	7	3
4	8	3	7	6	5	2	1	9
5	7	9	3	1	2	8	4	6
8	5	6	9	4	3	7	2	1
9	1	4	6	2	7	3	5	8
7	3	2	8	5	1	6	9	4
2	4	7	1	3	6	9	8	5
1	6	5	2	8	9	4	3	7
3	9	8	5	7	4	1	6	2

57

2	1	4	5	6	7	8	3	9
7	3	8	4	9	2	1	6	5
9	6	5	1	3	8	7	2	4
5	9	2	6	7	1	4	8	3
6	7	1	3	8	4	9	5	2
8	4	3	9	2	5	6	1	7
3	2	6	7	1	9	5	4	8
4	8	7	2	5	6	3	9	1
1	5	9	8	4	3	2	7	6

58

9	6	1	4	5	3	7	8	2
3	4	5	2	7	8	6	1	9
7	2	8	1	6	9	3	5	4
6	3	4	5	9	2	1	7	8
2	5	7	6	8	1	9	4	3
1	8	9	7	3	4	5	2	6
5	7	3	8	4	6	2	9	1
8	9	2	3	1	7	4	6	5
4	1	6	9	2	5	8	3	7

59

3	9	4	6	1	7	2	5	8
5	7	6	4	8	2	3	1	9
2	1	8	9	3	5	6	7	4
4	6	1	2	5	9	7	8	3
8	2	5	7	4	3	9	6	1
7	3	9	8	6	1	5	4	2
1	5	2	3	7	8	4	9	6
9	4	7	1	2	6	8	3	5
6	8	3	5	9	4	1	2	7

60

6	2	3	4	5	8	9	1	7
9	8	4	1	6	7	3	5	2
1	5	7	9	2	3	8	4	6
3	1	2	8	7	5	4	6	9
8	6	9	2	3	4	1	7	5
4	7	5	6	1	9	2	8	3
5	4	8	3	9	6	7	2	1
2	9	6	7	8	1	5	3	4
7	3	1	5	4	2	6	9	8

61

3	6	7	8	2	5	9	1	4
9	1	8	4	7	3	2	6	5
4	5	2	1	9	6	7	3	8
7	9	5	2	6	4	1	8	3
2	3	6	5	1	8	4	9	7
8	4	1	9	3	7	5	2	6
5	8	9	3	4	1	6	7	2
6	2	4	7	8	9	3	5	1
1	7	3	6	5	2	8	4	9

62

6	5	8	1	4	3	9	2	7
2	9	4	6	8	7	3	1	5
1	3	7	2	5	9	8	6	4
8	7	5	4	9	6	2	3	1
3	4	1	7	2	8	6	5	9
9	2	6	5	3	1	7	4	8
7	1	3	9	6	5	4	8	2
4	8	9	3	1	2	5	7	6
5	6	2	8	7	4	1	9	3

63

6	5	2	9	7	3	1	8	4
4	3	1	5	2	8	9	7	6
8	9	7	1	6	4	2	3	5
5	7	4	2	9	6	3	1	8
1	8	6	3	5	7	4	9	2
9	2	3	8	4	1	6	5	7
2	4	9	7	3	5	8	6	1
7	6	8	4	1	9	5	2	3
3	1	5	6	8	2	7	4	9

64

9	3	8	4	1	6	7	2	5
7	6	2	9	3	5	1	8	4
5	1	4	7	8	2	6	3	9
6	8	1	2	5	3	9	4	7
2	7	5	8	4	9	3	1	6
3	4	9	1	6	7	2	5	8
8	2	3	6	9	4	5	7	1
1	9	7	5	2	8	4	6	3
4	5	6	3	7	1	8	9	2

65

9	2	6	5	1	7	4	3	8
4	7	8	2	9	3	5	1	6
1	5	3	6	4	8	9	7	2
3	4	5	8	2	9	1	6	7
8	9	2	7	6	1	3	5	4
7	6	1	4	3	5	8	2	9
2	8	7	1	5	4	6	9	3
5	3	4	9	7	6	2	8	1
6	1	9	3	8	2	7	4	5

66

9	7	2	6	5	1	8	3	4
6	1	3	9	4	8	5	7	2
5	8	4	7	2	3	1	9	6
7	9	8	3	6	2	4	5	1
4	5	1	8	9	7	2	6	3
2	3	6	5	1	4	9	8	7
8	2	7	4	3	5	6	1	9
3	4	9	1	8	6	7	2	5
1	6	5	2	7	9	3	4	8

67

4	7	2	8	9	5	1	3	6
3	8	5	6	7	1	9	4	2
9	1	6	3	2	4	8	7	5
1	6	8	4	3	7	2	5	9
7	4	9	5	6	2	3	1	8
5	2	3	1	8	9	4	6	7
6	3	4	2	5	8	7	9	1
2	9	1	7	4	6	5	8	3
8	5	7	9	1	3	6	2	4

68

8	4	6	7	9	1	5	2	3
7	1	2	5	4	3	6	9	8
9	3	5	2	6	8	4	7	1
4	7	9	3	2	5	8	1	6
2	5	1	6	8	9	7	3	4
6	8	3	1	7	4	2	5	9
3	6	7	4	1	2	9	8	5
5	2	8	9	3	6	1	4	7
1	9	4	8	5	7	3	6	2

69

1	3	5	9	8	4	6	2	7
2	6	9	1	5	7	4	8	3
8	4	7	2	6	3	5	9	1
3	9	6	4	7	5	2	1	8
4	7	1	6	2	8	3	5	9
5	8	2	3	1	9	7	6	4
9	5	8	7	4	2	1	3	6
6	2	4	8	3	1	9	7	5
7	1	3	5	9	6	8	4	2

70

4	8	7	6	9	1	3	5	2
2	3	6	8	5	7	1	4	9
9	1	5	4	3	2	8	7	6
8	9	2	1	4	5	6	3	7
7	5	3	9	2	6	4	8	1
1	6	4	3	7	8	2	9	5
5	2	1	7	8	4	9	6	3
6	4	9	5	1	3	7	2	8
3	7	8	2	6	9	5	1	4

71

1	4	8	3	5	2	9	7	6
3	6	7	8	9	1	2	5	4
9	2	5	7	4	6	8	3	1
4	9	2	6	3	8	7	1	5
6	5	1	2	7	4	3	9	8
8	7	3	5	1	9	4	6	2
7	8	6	1	2	3	5	4	9
5	1	9	4	8	7	6	2	3
2	3	4	9	6	5	1	8	7

72

1	4	2	3	8	7	9	6	5
6	9	7	5	1	2	8	3	4
8	3	5	4	6	9	2	1	7
2	7	9	8	3	4	1	5	6
3	1	4	6	2	5	7	9	8
5	6	8	9	7	1	3	4	2
4	2	1	7	5	3	6	8	9
7	5	6	1	9	8	4	2	3
9	8	3	2	4	6	5	7	1

73

3	2	8	6	5	9	1	7	4
6	7	9	2	4	1	5	8	3
1	4	5	8	3	7	6	2	9
9	5	4	1	8	6	7	3	2
7	1	3	4	9	2	8	5	6
2	8	6	5	7	3	4	9	1
4	3	7	9	6	5	2	1	8
8	9	1	7	2	4	3	6	5
5	6	2	3	1	8	9	4	7

74

8	4	9	5	3	1	6	2	7
1	5	3	2	6	7	9	8	4
2	6	7	4	8	9	3	1	5
6	3	4	1	5	8	7	9	2
9	7	2	6	4	3	1	5	8
5	1	8	9	7	2	4	3	6
3	2	6	8	9	4	5	7	1
7	8	5	3	1	6	2	4	9
4	9	1	7	2	5	8	6	3

75

4	1	7	2	5	9	6	3	8
9	3	5	8	7	6	2	4	1
8	6	2	4	1	3	9	7	5
5	4	1	7	2	8	3	6	9
6	7	8	9	3	5	1	2	4
3	2	9	6	4	1	8	5	7
2	9	4	3	8	7	5	1	6
1	8	3	5	6	4	7	9	2
7	5	6	1	9	2	4	8	3

76

9	6	1	4	2	5	7	8	3
3	5	4	9	7	8	2	1	6
2	8	7	1	6	3	9	4	5
1	3	9	7	8	2	6	5	4
7	4	6	3	5	1	8	2	9
5	2	8	6	9	4	1	3	7
8	9	5	2	3	7	4	6	1
6	1	2	5	4	9	3	7	8
4	7	3	8	1	6	5	9	2

77

6	5	3	4	2	9	1	8	7
9	1	7	5	3	8	2	4	6
4	2	8	7	1	6	3	5	9
2	7	4	6	8	3	5	9	1
3	9	6	2	5	1	8	7	4
5	8	1	9	4	7	6	2	3
1	3	2	8	9	4	7	6	5
7	4	5	3	6	2	9	1	8
8	6	9	1	7	5	4	3	2

78

2	7	9	8	5	3	1	6	4
3	6	1	9	4	7	5	2	8
8	4	5	2	6	1	9	7	3
1	9	7	3	8	5	2	4	6
4	5	8	6	7	2	3	1	9
6	2	3	1	9	4	8	5	7
9	1	4	7	2	8	6	3	5
5	3	6	4	1	9	7	8	2
7	8	2	5	3	6	4	9	1

79

9	7	4	1	6	5	8	2	3
5	6	2	3	8	4	9	7	1
1	3	8	2	7	9	6	4	5
7	4	1	8	3	6	2	5	9
3	8	9	5	2	1	4	6	7
2	5	6	9	4	7	3	1	8
6	9	3	7	1	2	5	8	4
8	2	7	4	5	3	1	9	6
4	1	5	6	9	8	7	3	2

80

4	3	7	9	8	5	1	2	6
2	9	1	6	7	3	4	5	8
5	6	8	1	4	2	9	7	3
3	7	5	8	2	9	6	4	1
6	8	4	5	1	7	2	3	9
1	2	9	3	6	4	5	8	7
9	1	3	2	5	8	7	6	4
7	5	6	4	3	1	8	9	2
8	4	2	7	9	6	3	1	5

81

8	6	2	5	3	4	1	9	7
4	5	3	9	1	7	2	6	8
7	1	9	6	8	2	5	4	3
5	7	4	1	2	6	3	8	9
3	2	8	4	5	9	6	7	1
6	9	1	3	7	8	4	5	2
1	8	6	2	9	5	7	3	4
2	4	7	8	6	3	9	1	5
9	3	5	7	4	1	8	2	6

82

7	3	5	6	4	2	9	8	1
2	9	8	3	5	1	4	7	6
1	4	6	8	7	9	5	3	2
4	2	9	7	1	6	8	5	3
6	7	3	2	8	5	1	4	9
8	5	1	9	3	4	2	6	7
5	6	7	1	2	8	3	9	4
9	1	4	5	6	3	7	2	8
3	8	2	4	9	7	6	1	5

83

6	3	2	5	7	1	8	9	4
4	5	1	8	2	9	7	3	6
9	7	8	3	6	4	5	1	2
7	9	6	1	4	2	3	5	8
2	1	3	7	5	8	4	6	9
8	4	5	9	3	6	2	7	1
5	6	9	4	8	7	1	2	3
1	8	7	2	9	3	6	4	5
3	2	4	6	1	5	9	8	7

84

3	5	8	6	9	7	2	4	1
1	9	2	4	8	5	6	7	3
6	7	4	2	1	3	5	9	8
5	3	9	8	7	1	4	6	2
8	6	7	3	4	2	9	1	5
4	2	1	9	5	6	8	3	7
2	1	3	5	6	9	7	8	4
7	4	6	1	2	8	3	5	9
9	8	5	7	3	4	1	2	6

85

2	4	5	8	9	1	6	7	3
7	6	1	5	3	4	8	9	2
8	9	3	6	2	7	1	5	4
4	3	8	7	5	2	9	1	6
6	5	9	4	1	8	3	2	7
1	7	2	9	6	3	5	4	8
5	8	4	1	7	6	2	3	9
3	1	6	2	4	9	7	8	5
9	2	7	3	8	5	4	6	1

86

4	7	1	6	8	9	5	2	3
3	9	6	2	7	5	8	1	4
8	2	5	4	1	3	7	9	6
1	3	2	7	9	4	6	8	5
7	5	8	3	2	6	9	4	1
6	4	9	1	5	8	3	7	2
2	1	3	9	6	7	4	5	8
9	8	4	5	3	1	2	6	7
5	6	7	8	4	2	1	3	9

87

8	6	7	1	5	4	3	9	2
5	9	1	2	6	3	7	4	8
2	4	3	9	7	8	1	5	6
1	8	4	6	2	7	5	3	9
6	3	5	4	8	9	2	7	1
7	2	9	3	1	5	8	6	4
9	1	2	7	3	6	4	8	5
4	7	8	5	9	1	6	2	3
3	5	6	8	4	2	9	1	7

88

3	4	7	2	1	9	8	6	5
5	8	6	4	3	7	1	2	9
2	1	9	5	6	8	7	4	3
8	9	2	6	5	4	3	1	7
1	7	5	3	9	2	4	8	6
6	3	4	7	8	1	9	5	2
4	5	8	9	2	3	6	7	1
9	6	1	8	7	5	2	3	4
7	2	3	1	4	6	5	9	8

89

9	8	4	1	6	2	7	3	5
7	5	6	8	9	3	1	2	4
1	2	3	5	7	4	8	6	9
3	7	8	2	5	1	4	9	6
4	6	5	3	8	9	2	1	7
2	9	1	7	4	6	3	5	8
6	4	2	9	3	7	5	8	1
5	1	9	4	2	8	6	7	3
8	3	7	6	1	5	9	4	2

90

3	2	8	4	6	1	9	5	7
5	1	9	3	7	2	6	4	8
4	6	7	9	8	5	3	2	1
2	9	1	7	3	8	5	6	4
8	4	5	2	1	6	7	3	9
7	3	6	5	9	4	1	8	2
9	5	2	6	4	7	8	1	3
6	8	3	1	2	9	4	7	5
1	7	4	8	5	3	2	9	6

91

6	5	3	4	1	2	7	8	9
9	7	8	5	3	6	2	1	4
2	4	1	9	8	7	3	6	5
8	9	4	2	7	3	1	5	6
3	1	7	6	9	5	4	2	8
5	2	6	8	4	1	9	3	7
7	8	2	3	6	4	5	9	1
4	3	9	1	5	8	6	7	2
1	6	5	7	2	9	8	4	3

92

9	5	6	8	2	4	7	1	3
2	1	8	9	3	7	4	6	5
4	7	3	5	1	6	8	2	9
7	8	4	1	5	2	3	9	6
5	3	9	6	7	8	1	4	2
6	2	1	3	4	9	5	7	8
3	6	2	4	8	1	9	5	7
8	4	7	2	9	5	6	3	1
1	9	5	7	6	3	2	8	4

93

3	6	2	4	9	8	5	1	7
4	1	9	7	5	3	2	6	8
5	8	7	6	2	1	4	3	9
8	2	6	5	4	7	1	9	3
9	5	3	1	6	2	7	8	4
7	4	1	8	3	9	6	2	5
1	3	8	2	7	5	9	4	6
2	7	4	9	8	6	3	5	1
6	9	5	3	1	4	8	7	2

94

6	8	2	1	4	9	7	3	5
1	9	5	2	3	7	4	6	8
3	7	4	5	8	6	9	1	2
7	2	1	4	5	8	6	9	3
4	3	8	6	9	2	5	7	1
5	6	9	7	1	3	2	8	4
2	1	6	8	7	5	3	4	9
8	5	3	9	6	4	1	2	7
9	4	7	3	2	1	8	5	6

95

5	2	9	7	1	4	3	6	8
4	3	1	6	9	8	7	5	2
6	8	7	3	5	2	1	9	4
9	1	8	4	7	6	2	3	5
3	6	4	8	2	5	9	7	1
7	5	2	1	3	9	4	8	6
8	7	5	9	4	1	6	2	3
2	4	3	5	6	7	8	1	9
1	9	6	2	8	3	5	4	7

96

6	1	9	7	5	8	2	4	3
5	4	8	6	3	2	7	1	9
3	2	7	9	4	1	5	8	6
4	8	1	2	9	3	6	5	7
7	6	3	5	8	4	1	9	2
2	9	5	1	6	7	4	3	8
1	3	6	4	2	9	8	7	5
9	7	2	8	1	5	3	6	4
8	5	4	3	7	6	9	2	1

97

3	2	6	1	4	5	9	8	7
8	4	7	9	6	3	2	1	5
1	5	9	2	8	7	4	6	3
6	1	4	8	5	9	3	7	2
2	8	3	6	7	4	1	5	9
9	7	5	3	2	1	8	4	6
4	3	2	7	1	6	5	9	8
7	9	1	5	3	8	6	2	4
5	6	8	4	9	2	7	3	1

98

6	3	1	8	9	2	7	5	4
5	8	7	1	4	3	9	6	2
2	4	9	7	6	5	1	3	8
3	6	5	9	1	8	2	4	7
7	1	8	5	2	4	3	9	6
9	2	4	3	7	6	8	1	5
1	7	6	2	5	9	4	8	3
4	9	3	6	8	7	5	2	1
8	5	2	4	3	1	6	7	9

99

6	9	1	5	7	4	3	2	8
4	7	2	8	6	3	5	1	9
8	5	3	1	2	9	7	6	4
7	1	8	2	4	5	6	9	3
3	2	6	9	1	8	4	7	5
5	4	9	6	3	7	2	8	1
9	3	7	4	8	2	1	5	6
1	8	4	7	5	6	9	3	2
2	6	5	3	9	1	8	4	7

100

9	4	6	7	5	8	1	2	3
1	3	8	6	4	2	5	9	7
5	2	7	9	1	3	4	8	6
4	1	9	2	8	6	7	3	5
6	7	2	3	9	5	8	4	1
3	8	5	1	7	4	9	6	2
8	6	1	4	3	7	2	5	9
2	9	4	5	6	1	3	7	8
7	5	3	8	2	9	6	1	4

101

7	1	8	6	2	4	9	3	5
9	2	4	3	1	5	6	7	8
5	3	6	7	9	8	4	1	2
6	5	3	2	4	7	8	9	1
1	7	2	8	6	9	5	4	3
8	4	9	5	3	1	2	6	7
4	8	5	9	7	3	1	2	6
2	9	7	1	8	6	3	5	4
3	6	1	4	5	2	7	8	9

102

4	2	8	3	5	6	7	1	9
5	9	1	8	4	7	3	6	2
6	3	7	9	2	1	4	5	8
3	8	6	2	7	5	9	4	1
9	7	4	6	1	8	5	2	3
1	5	2	4	3	9	8	7	6
8	4	9	5	6	2	1	3	7
2	1	5	7	9	3	6	8	4
7	6	3	1	8	4	2	9	5

103

3	2	6	7	1	9	8	4	5
8	7	5	3	6	4	1	9	2
9	1	4	8	5	2	6	7	3
4	9	8	6	3	5	2	1	7
7	6	3	2	9	1	5	8	4
1	5	2	4	8	7	9	3	6
5	4	1	9	7	6	3	2	8
6	8	7	1	2	3	4	5	9
2	3	9	5	4	8	7	6	1

104

5	1	3	4	9	7	6	8	2
7	9	8	1	2	6	4	3	5
4	2	6	5	8	3	7	9	1
8	5	2	9	6	1	3	4	7
9	7	1	3	5	4	2	6	8
6	3	4	2	7	8	5	1	9
3	4	9	7	1	5	8	2	6
2	6	5	8	4	9	1	7	3
1	8	7	6	3	2	9	5	4

105

8	3	6	5	1	9	2	7	4
1	5	2	8	7	4	9	6	3
4	7	9	3	2	6	5	8	1
7	8	4	6	9	3	1	5	2
5	2	1	7	4	8	3	9	6
6	9	3	1	5	2	7	4	8
3	4	7	2	8	5	6	1	9
9	6	5	4	3	1	8	2	7
2	1	8	9	6	7	4	3	5

106

2	3	7	9	4	5	6	1	8
6	1	5	7	8	3	4	9	2
9	4	8	6	2	1	7	3	5
4	7	6	2	9	8	3	5	1
1	9	2	3	5	7	8	4	6
5	8	3	4	1	6	2	7	9
8	6	4	1	7	9	5	2	3
3	2	1	5	6	4	9	8	7
7	5	9	8	3	2	1	6	4

107

8	6	2	5	4	7	9	1	3
9	1	3	8	6	2	7	4	5
7	4	5	3	9	1	6	8	2
5	2	8	9	7	4	1	3	6
1	7	6	2	8	3	5	9	4
4	3	9	1	5	6	8	2	7
3	8	1	7	2	5	4	6	9
2	5	4	6	1	9	3	7	8
6	9	7	4	3	8	2	5	1

108

3	8	2	5	1	6	9	4	7
9	5	1	7	2	4	3	6	8
6	4	7	9	8	3	1	5	2
2	9	6	4	5	7	8	1	3
8	1	5	2	3	9	6	7	4
4	7	3	1	6	8	5	2	9
7	2	8	6	9	1	4	3	5
1	3	4	8	7	5	2	9	6
5	6	9	3	4	2	7	8	1

109

3	1	5	9	8	7	4	6	2
8	6	7	3	4	2	9	5	1
9	2	4	6	1	5	3	7	8
7	4	6	1	9	3	8	2	5
2	5	9	7	6	8	1	3	4
1	3	8	5	2	4	6	9	7
4	8	3	2	7	6	5	1	9
5	9	2	8	3	1	7	4	6
6	7	1	4	5	9	2	8	3

110

9	4	2	5	6	1	7	3	8
6	1	3	2	8	7	5	9	4
7	5	8	3	4	9	6	1	2
1	6	7	9	5	4	2	8	3
5	2	4	8	1	3	9	6	7
8	3	9	6	7	2	1	4	5
2	9	1	4	3	5	8	7	6
3	8	5	7	9	6	4	2	1
4	7	6	1	2	8	3	5	9

111

8	4	7	6	5	1	3	2	9
5	3	2	8	9	4	1	6	7
9	1	6	7	3	2	5	4	8
7	5	1	9	2	3	4	8	6
3	9	4	1	6	8	2	7	5
2	6	8	4	7	5	9	3	1
4	8	9	3	1	7	6	5	2
1	2	3	5	8	6	7	9	4
6	7	5	2	4	9	8	1	3

112

7	3	4	5	8	2	6	9	1
9	1	5	7	4	6	3	8	2
2	8	6	3	1	9	4	5	7
4	2	9	1	3	7	8	6	5
6	5	8	9	2	4	1	7	3
1	7	3	8	6	5	9	2	4
5	4	1	2	9	8	7	3	6
8	6	2	4	7	3	5	1	9
3	9	7	6	5	1	2	4	8

113

8	4	3	2	1	9	5	6	7
5	7	6	8	4	3	9	2	1
9	1	2	6	5	7	3	4	8
3	8	1	9	2	4	7	5	6
2	9	5	1	7	6	8	3	4
4	6	7	5	3	8	2	1	9
6	2	4	7	8	5	1	9	3
7	5	9	3	6	1	4	8	2
1	3	8	4	9	2	6	7	5

114

7	8	3	2	6	1	5	4	9
6	5	1	9	4	3	7	2	8
2	9	4	7	5	8	1	3	6
1	7	6	3	9	2	8	5	4
3	4	9	8	7	5	6	1	2
8	2	5	4	1	6	3	9	7
5	3	8	6	2	9	4	7	1
9	1	7	5	8	4	2	6	3
4	6	2	1	3	7	9	8	5

115

2	4	7	8	9	3	5	6	1
8	5	3	6	7	1	9	2	4
6	1	9	4	2	5	8	3	7
7	2	1	9	4	6	3	8	5
9	8	6	5	3	7	1	4	2
4	3	5	1	8	2	6	7	9
5	6	8	2	1	4	7	9	3
1	7	2	3	6	9	4	5	8
3	9	4	7	5	8	2	1	6

116

9	2	5	4	6	3	8	1	7
3	6	8	2	7	1	4	5	9
7	4	1	5	8	9	2	6	3
4	1	9	6	3	2	7	8	5
5	7	6	9	1	8	3	2	4
2	8	3	7	4	5	6	9	1
8	3	7	1	9	6	5	4	2
6	9	2	3	5	4	1	7	8
1	5	4	8	2	7	9	3	6

117

5	3	6	9	2	8	7	4	1
2	8	4	6	7	1	3	9	5
7	1	9	3	4	5	6	8	2
8	6	3	4	5	2	1	7	9
9	7	2	8	1	3	5	6	4
4	5	1	7	6	9	2	3	8
6	9	5	2	8	7	4	1	3
1	4	8	5	3	6	9	2	7
3	2	7	1	9	4	8	5	6

118

1	9	4	5	6	2	3	8	7
6	3	7	1	4	8	5	2	9
5	8	2	3	7	9	4	1	6
8	2	1	7	5	6	9	4	3
4	5	3	9	8	1	6	7	2
9	7	6	2	3	4	8	5	1
3	1	9	8	2	5	7	6	4
7	4	8	6	1	3	2	9	5
2	6	5	4	9	7	1	3	8

119

8	9	5	2	4	3	6	7	1
1	3	6	9	5	7	2	4	8
2	7	4	8	6	1	5	9	3
6	2	9	5	1	4	3	8	7
3	4	7	6	9	8	1	2	5
5	8	1	7	3	2	4	6	9
9	5	2	1	7	6	8	3	4
7	6	3	4	8	5	9	1	2
4	1	8	3	2	9	7	5	6

120

6	9	2	7	3	5	4	1	8
8	5	3	6	1	4	7	2	9
1	4	7	8	2	9	5	3	6
9	6	8	5	7	3	1	4	2
5	2	4	9	8	1	3	6	7
7	3	1	4	6	2	8	9	5
4	8	6	3	9	7	2	5	1
2	7	5	1	4	6	9	8	3
3	1	9	2	5	8	6	7	4

121

9	7	5	4	1	2	8	3	6
1	8	3	6	5	7	2	9	4
6	4	2	9	8	3	7	5	1
8	9	4	1	6	5	3	2	7
2	5	7	8	3	4	1	6	9
3	1	6	2	7	9	4	8	5
4	3	8	7	9	6	5	1	2
5	2	9	3	4	1	6	7	8
7	6	1	5	2	8	9	4	3

122

9	7	6	2	5	1	4	3	8
3	1	8	6	4	7	5	9	2
4	2	5	3	9	8	6	1	7
5	6	1	4	7	9	8	2	3
8	9	7	5	2	3	1	4	6
2	4	3	8	1	6	7	5	9
1	3	9	7	8	4	2	6	5
6	8	2	1	3	5	9	7	4
7	5	4	9	6	2	3	8	1

123

3	7	2	9	8	5	6	4	1
9	4	6	7	3	1	8	5	2
8	1	5	2	6	4	7	9	3
2	6	3	4	9	7	1	8	5
1	9	8	5	2	3	4	6	7
4	5	7	8	1	6	2	3	9
7	2	4	6	5	9	3	1	8
6	3	9	1	7	8	5	2	4
5	8	1	3	4	2	9	7	6

124

4	5	3	6	8	7	9	1	2
7	6	8	1	9	2	4	3	5
1	9	2	5	4	3	7	6	8
9	7	5	8	3	4	6	2	1
2	4	1	9	6	5	3	8	7
3	8	6	2	7	1	5	9	4
8	3	4	7	2	9	1	5	6
6	1	9	4	5	8	2	7	3
5	2	7	3	1	6	8	4	9

125

2	3	9	5	8	4	1	7	6
5	8	6	2	1	7	4	3	9
4	1	7	3	9	6	5	8	2
6	2	4	9	5	8	7	1	3
3	5	1	6	7	2	8	9	4
7	9	8	1	4	3	2	6	5
9	7	5	4	3	1	6	2	8
1	4	2	8	6	9	3	5	7
8	6	3	7	2	5	9	4	1

126

7	8	1	5	3	4	9	2	6
4	5	6	7	9	2	1	3	8
3	9	2	1	6	8	5	7	4
6	2	7	4	8	5	3	1	9
5	1	3	9	2	6	8	4	7
8	4	9	3	7	1	2	6	5
9	6	8	2	4	3	7	5	1
1	3	4	8	5	7	6	9	2
2	7	5	6	1	9	4	8	3

127

9	8	4	7	1	6	2	3	5
7	5	3	2	9	4	8	6	1
2	6	1	8	3	5	4	7	9
5	2	6	9	8	3	1	4	7
3	1	9	5	4	7	6	2	8
4	7	8	6	2	1	5	9	3
1	4	5	3	6	9	7	8	2
8	9	7	4	5	2	3	1	6
6	3	2	1	7	8	9	5	4

128

1	9	4	2	8	3	5	6	7
3	5	7	6	1	4	8	9	2
6	8	2	9	5	7	4	3	1
2	7	8	4	6	9	3	1	5
4	1	3	8	7	5	9	2	6
9	6	5	3	2	1	7	8	4
7	4	6	1	3	8	2	5	9
8	2	9	5	4	6	1	7	3
5	3	1	7	9	2	6	4	8

129

9	4	6	2	7	8	3	5	1
7	8	1	4	5	3	2	6	9
2	5	3	9	1	6	7	8	4
4	3	7	1	6	5	8	9	2
5	9	8	7	4	2	6	1	3
6	1	2	8	3	9	4	7	5
8	6	4	3	9	1	5	2	7
3	2	9	5	8	7	1	4	6
1	7	5	6	2	4	9	3	8

130

6	5	4	2	1	8	3	9	7
1	7	9	5	6	3	2	4	8
8	2	3	4	9	7	5	6	1
7	6	1	3	2	5	4	8	9
4	9	5	6	8	1	7	2	3
3	8	2	7	4	9	1	5	6
5	3	8	9	7	4	6	1	2
2	1	7	8	5	6	9	3	4
9	4	6	1	3	2	8	7	5

131

2	1	8	4	9	7	3	5	6
4	9	3	5	1	6	7	8	2
5	6	7	2	8	3	1	9	4
9	8	4	3	5	1	6	2	7
6	2	1	7	4	8	9	3	5
7	3	5	9	6	2	4	1	8
3	5	2	1	7	4	8	6	9
1	7	6	8	2	9	5	4	3
8	4	9	6	3	5	2	7	1

132

1	7	3	2	6	8	5	4	9
9	4	6	7	3	5	1	8	2
2	8	5	4	9	1	6	3	7
3	2	4	8	5	7	9	6	1
7	1	9	3	4	6	8	2	5
6	5	8	1	2	9	4	7	3
4	3	1	5	8	2	7	9	6
5	9	2	6	7	4	3	1	8
8	6	7	9	1	3	2	5	4

133

7	1	9	2	6	4	5	8	3
4	8	3	7	5	1	2	9	6
5	6	2	9	8	3	1	7	4
6	9	4	3	1	2	7	5	8
1	5	8	4	7	6	3	2	9
2	3	7	8	9	5	6	4	1
8	7	1	5	3	9	4	6	2
3	4	5	6	2	8	9	1	7
9	2	6	1	4	7	8	3	5

134

2	3	1	7	8	6	5	9	4
5	6	4	2	3	9	1	8	7
8	9	7	4	1	5	3	2	6
9	8	6	3	7	2	4	5	1
1	5	3	8	6	4	9	7	2
4	7	2	5	9	1	6	3	8
7	1	5	6	2	3	8	4	9
3	2	9	1	4	8	7	6	5
6	4	8	9	5	7	2	1	3

135

7	6	8	4	2	5	1	9	3
4	9	2	3	8	1	6	5	7
5	1	3	6	9	7	4	2	8
3	2	6	8	7	4	5	1	9
9	4	1	5	3	6	7	8	2
8	5	7	9	1	2	3	4	6
2	8	4	1	6	3	9	7	5
1	3	9	7	5	8	2	6	4
6	7	5	2	4	9	8	3	1

136

5	2	9	4	7	8	3	6	1
3	8	4	2	1	6	5	7	9
6	1	7	5	3	9	4	2	8
2	4	8	1	9	3	7	5	6
7	9	3	6	4	5	1	8	2
1	6	5	7	8	2	9	4	3
4	3	2	9	6	7	8	1	5
9	5	1	8	2	4	6	3	7
8	7	6	3	5	1	2	9	4

137

6	7	9	2	1	5	4	8	3
4	2	5	9	3	8	6	1	7
8	3	1	7	4	6	5	9	2
7	5	2	6	9	4	1	3	8
3	9	4	1	8	7	2	6	5
1	8	6	3	5	2	7	4	9
9	4	7	8	2	1	3	5	6
5	6	8	4	7	3	9	2	1
2	1	3	5	6	9	8	7	4

138

7	9	8	6	1	5	2	3	4
3	4	1	7	8	2	9	6	5
5	6	2	9	4	3	1	8	7
4	1	6	2	5	7	3	9	8
8	5	9	3	6	1	7	4	2
2	7	3	8	9	4	5	1	6
1	2	4	5	3	6	8	7	9
9	3	7	4	2	8	6	5	1
6	8	5	1	7	9	4	2	3

139

1	8	3	4	9	7	5	6	2
4	5	7	6	2	8	3	1	9
6	9	2	1	3	5	4	8	7
2	7	5	9	1	4	8	3	6
3	6	4	7	8	2	9	5	1
8	1	9	5	6	3	7	2	4
9	3	1	8	7	6	2	4	5
7	4	8	2	5	1	6	9	3
5	2	6	3	4	9	1	7	8

140

8	9	5	1	7	3	2	6	4
1	4	2	9	8	6	3	5	7
6	3	7	5	4	2	8	1	9
4	6	8	2	9	5	7	3	1
5	1	9	3	6	7	4	2	8
7	2	3	8	1	4	5	9	6
2	8	6	7	5	9	1	4	3
3	7	4	6	2	1	9	8	5
9	5	1	4	3	8	6	7	2

141

3	8	7	5	9	2	4	1	6
4	5	1	7	8	6	9	2	3
9	6	2	4	1	3	8	5	7
8	4	3	1	5	7	2	6	9
1	9	6	3	2	8	5	7	4
7	2	5	6	4	9	1	3	8
2	3	9	8	7	5	6	4	1
5	7	4	9	6	1	3	8	2
6	1	8	2	3	4	7	9	5

142

3	1	9	8	6	4	2	7	5
8	7	2	9	3	5	6	4	1
4	6	5	2	1	7	8	9	3
2	3	6	4	9	1	7	5	8
1	9	7	3	5	8	4	2	6
5	8	4	6	7	2	1	3	9
9	2	8	5	4	6	3	1	7
7	4	3	1	8	9	5	6	2
6	5	1	7	2	3	9	8	4

143

7	3	2	5	9	1	6	8	4
5	9	6	4	3	8	7	1	2
8	1	4	7	2	6	5	9	3
3	7	5	9	8	4	2	6	1
6	4	8	3	1	2	9	7	5
9	2	1	6	7	5	3	4	8
4	8	7	2	5	9	1	3	6
1	5	3	8	6	7	4	2	9
2	6	9	1	4	3	8	5	7

144

4	6	2	3	5	1	7	9	8
1	9	5	2	7	8	3	6	4
3	7	8	9	6	4	1	5	2
9	1	6	7	2	3	4	8	5
8	2	7	6	4	5	9	3	1
5	3	4	1	8	9	6	2	7
6	4	3	5	1	2	8	7	9
7	5	1	8	9	6	2	4	3
2	8	9	4	3	7	5	1	6

145

9	5	1	2	4	8	3	7	6
3	6	2	1	9	7	4	8	5
4	8	7	6	5	3	9	2	1
1	9	6	7	3	2	8	5	4
7	2	8	4	1	5	6	3	9
5	3	4	8	6	9	7	1	2
2	1	3	9	7	4	5	6	8
6	7	9	5	8	1	2	4	3
8	4	5	3	2	6	1	9	7

146

9	8	5	3	7	6	2	1	4
6	2	1	9	4	5	8	7	3
3	4	7	2	1	8	6	5	9
4	5	2	8	3	7	9	6	1
1	9	3	5	6	4	7	8	2
7	6	8	1	2	9	3	4	5
2	3	6	4	8	1	5	9	7
5	7	4	6	9	3	1	2	8
8	1	9	7	5	2	4	3	6

147

3	1	9	2	4	5	8	7	6
4	5	7	9	8	6	1	2	3
2	6	8	1	3	7	4	5	9
8	3	6	7	2	9	5	1	4
9	7	1	6	5	4	3	8	2
5	4	2	3	1	8	9	6	7
1	2	4	5	7	3	6	9	8
7	9	3	8	6	1	2	4	5
6	8	5	4	9	2	7	3	1

148

8	7	4	6	2	9	3	5	1
3	1	5	8	4	7	9	6	2
6	2	9	3	5	1	4	7	8
7	8	6	5	3	2	1	4	9
9	4	3	1	6	8	7	2	5
2	5	1	7	9	4	6	8	3
1	9	2	4	8	6	5	3	7
4	3	7	2	1	5	8	9	6
5	6	8	9	7	3	2	1	4

149

4	5	3	1	7	8	9	2	6
2	8	7	6	3	9	1	5	4
6	1	9	2	5	4	7	8	3
3	2	4	7	6	1	5	9	8
7	9	5	8	2	3	6	4	1
8	6	1	4	9	5	2	3	7
5	3	6	9	4	7	8	1	2
9	7	8	3	1	2	4	6	5
1	4	2	5	8	6	3	7	9

150

6	8	2	3	5	4	9	7	1
9	4	1	2	6	7	8	5	3
5	7	3	9	1	8	2	6	4
1	3	6	7	8	5	4	2	9
7	2	4	1	9	3	6	8	5
8	5	9	4	2	6	3	1	7
4	1	5	8	3	2	7	9	6
3	9	8	6	7	1	5	4	2
2	6	7	5	4	9	1	3	8

THE  TIMES

Su Doku

Also available:

<i>The Times Su Doku Book 1</i>	978-0-00-720732-9
<i>The Times Su Doku Book 2</i>	978-0-00-721350-4
<i>The Times Su Doku Book 3</i>	978-0-00-721426-6
<i>The Times Su Doku Book 4</i>	978-0-00-722241-4
<i>The Times Su Doku Book 5</i>	978-0-00-722242-1
<i>The Times Su Doku Book 6</i>	978-0-00-724753-0
<i>The Times Su Doku Book 7</i>	978-0-00-725040-0
<i>The Times Su Doku Book 8</i>	978-0-00-727257-0
<i>Times Su Doku for Beginners</i>	978-0-00-722598-9
<i>The Times Easy Su Doku</i>	978-0-00-723251-2
<i>The Times Difficult Su Doku Book 1</i>	978-0-00-723252-9
<i>The Times Difficult Su Doku Book 2</i>	978-0-00-730738-8
<i>The Times Fiendish Su Doku Book 1</i>	978-0-00-723253-6
<i>The Times Fiendish Su Doku Book 2</i>	978-0-00-730736-4

THE  TIMES

Su Doku

The Times Killer Su Doku Book 1

978-0-00-722363-3

The Times Killer Su Doku Book 2

978-0-00-723617-6

The Times Killer Su Doku Book 3

978-0-00-724800-1

The Times Killer Su Doku Book 4

978-0-00-727258-7

The Times Gem Su Doku Book 1

978-0-00-722588-0

The Times Gem Su Doku Book 2

978-0-00-726959-4

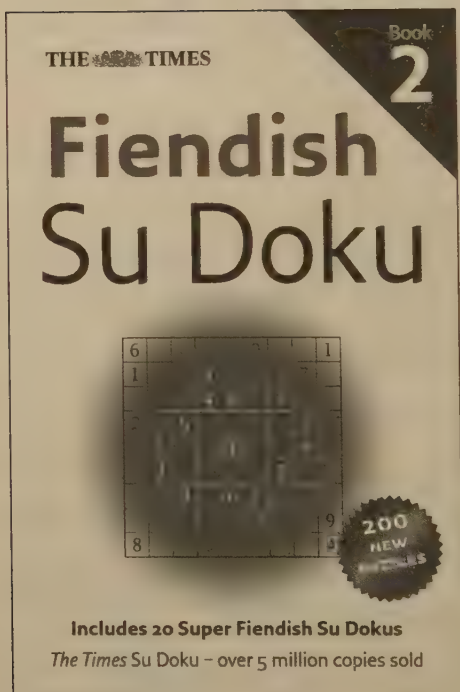
The Times Samurai Su Doku

978-0-00-724165-1

The Times Samurai Su Doku Book 2

978-0-00-725041-7

Also available



The Times Fiendish Su Doku Book 2
ISBN 978-0-00-730736-4

*If you enjoyed this book, try
KenKen, the successor to Su Doku*



The Times KenKen 1
ISBN 978-0-00-728824-3



The Times KenKen 2
ISBN 978-0-00-729090-1

System

0777613 6987

数

the

the

救獨

the

thai
thai

Shu.

then, then

The Su Doku in this book are provided by:

SUDOKUSOLVER.COM

Generate and solve your Su Doku for free.

SUDOKU

GODOKU

SAMURAI SUDOKU

SUPER SUDOKU

KILLER SUDOKU

Sign up for your own account with the following features:

- ▶ create your own exclusive Su Doku, Samurai Su Doku and Killer Su Doku puzzles
- ▶ solve and generate; then print or email any Su Doku puzzle
- ▶ solve popular puzzles instantly without entering any numbers
- ▶ view simple explanations of common and advanced Su Doku solving techniques
- ▶ free entry to win prizes with our Su Doku competitions

Enjoy playing Su Doku more at sudokusolver.com!

07553186270

THE TIMES

To help satisfy the nation's appetite for Su Doku, *The Times* offers up its ninth instalment of the Japanese puzzle.

Book 9 contains 150 previously unpublished puzzles, ranging from the mercifully Mild to the utterly unforgiving Super Fiendish.

It's getting tougher all the time, so be prepared to raise your game and wrangle once again with the devilish number grid.

TIMES BOOKS

£5.99 CAN\$9.99

ISBN 978-0-00-730584-1



www.collins.co.uk



Sustainable reading

www.harpercollins.co.uk/green

FSC + HarperCollins

Your choice makes a difference